A Radiometric and Color Balancing Method for Aerial Imagery Based on Remote Sensing Radiometric References

Yuting Qin, Zhiling Yu, and Yansong Duan

***Abstract*—Large-scale aerial imagery color consistency and fidelity are critical for accurate interpretation and analysis in applications such as surveying and mapping, ecological monitoring, and disaster assessment. However, existing techniques face two primary challenges: detail loss and local color casts in single-image correction due to the lack of a physical radiation model, and residual discrepancies in multi-image global optimization caused by insufficient radiometric references. To address these issues, this study proposes a radiometric and color balancing method for aerial imagery based on remote sensing radiometric references. The framework first constructs a physical model for nonuniform color compensation using radiometric control points (RCPs), effectively mitigating detail loss and local casts in individual images. Subsequently, radiometric constraints are integrated into the aerotriangulation framework to ensure global consistency across multiple images. Comprehensive evaluations on three representative aerial datasets demonstrate that the proposed method outperforms established approaches, including Wallis, Retinex, hybrid histogram matching (HHM), LORACCS, DCC, C4RSD, and RS-NormGAN. Experimental results indicate significant improvements, particularly in complex scenarios with uneven illumination. Compared to the top-performing benchmark, the proposed method reduced color distance (CD) and DeltaE by 36.36% and 28.91%, respectively, on the DMC dataset. For the DMZ1 dataset, which exhibits severe color casts, CD and DeltaE decreased by 8.06% and 28.32%, while peak signal-to-noise ratio (PSNR) increased by 0.21 dB. In the DMZ2 dataset, characterized by large illumination differences, mean squared error (mse) and DeltaE were reduced by 8.77% and 9.09%, respectively, with a PSNR improvement of 0.37 dB. Furthermore, the method excelled in preserving radiometric fidelity and texture details, offering a robust solution for high-quality remote sensing product generation.**

***Index Terms*—Aerial imagery, color consistency, physical radiometric model, radiometric aerotriangulation, radiometric and color balancing.**

I. Introduction

IN RECENT years, aerial imagery has been extensively utilized in fields such as cartographic mapping, geological disaster monitoring, and ecological environmental assessment, owing to its flexible acquisition modes, relatively low cost, and high spatiotemporal resolution. The clarity of aerial imagery is pivotal for extracting ground object information. However, factors such as variations in illumination conditions, terrain undulation, nonlinear sensor responses, and changes in camera attitude often lead to color inconsistencies and illumination nonuniformity within or across images. If not properly addressed, these issues lead to pronounced color and brightness discrepancies in mosaicked images, thereby compromising the quality of the final digital orthophoto maps (DOMs) and negatively impacting subsequent tasks such as image classification, object detection, and change detection. Consequently, there is an urgent demand for the development of high-fidelity radiometric and color balancing techniques to enhance the quality of aerial remote sensing products.

At the single-image level, issues typically manifest as uneven illumination distribution, such as vignetting, local overexposure, or underexposure. To mitigate these effects, existing mainstream methods are primarily based on illumination-reflectance models or adaptive statistical transformations. Although these methods achieve visual balance to a certain extent by decomposing images or adjusting global/local statistical properties, most of them are based on statistical features or simplified physical assumptions. They lack precise modeling of the actual radiometric physical processes of the scene. Consequently, they are prone to introducing side effects such as detail loss, texture attenuation, and color distortions. Particularly under complex illumination conditions, these approaches often struggle to balance local detail preservation with global uniformity.

At the multi-image mosaicking level, global color inconsistency across images is another prevalent issue. To eliminate color discrepancies during mosaicking, methods based on overlapping area constraints and global energy optimization are widely applied. These methods primarily focus on statistical relationship matching or visual consistency constraints. While these methods enhance the overall consistency of multiple images, they either suffer from error accumulation during local sequential propagation or fail to accurately characterize local nonlinear radiometric distortions due to the use of unified globalor block-based mapping models, leading to residual local color discrepancies. Therefore, there is a critical need to introduce reliable radiometric physical references into the global optimization framework, using physical constraints to guide statistical and visual optimization to overcome the bottleneck of incomplete local correction.

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The authors are with the School of Remote Sensing and Information Engineering, Wuhan University, Wuhan 430079, China (e-mail: qinyuting@whu.edu.cn; yuzhiling@whu.edu.cn; ysduan@whu.edu.cn).

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In general, current radiometric and color balancing methods encounter two primary challenges in addressing complex nonuniform color correction in aerial imagery: first, detail loss and local color casts in single-image correction due to the absence of a physical radiometric model; and second, residual local color discrepancies and insufficient physical consistency in multi-image global optimization due to the lack of a physical reference. To address these limitations, this article proposes a radiometric and color balancing method for aerial imagery based on remote sensing radiometric references. The main innovations of this article are as follows.

1) A physical radiometric model for nonuniformity color compensation is proposed. By describing the radiometric distortions process during imaging, this model provides a physical basis for correcting intraimage nonuniformity, effectively mitigating detail loss and local color casts.
2) A color consistency correction model based on radiometric aerotriangulation is introduced. By integrating radiometric reference information into the aerotriangulation adjustment framework, the model jointly optimizes the radiometric consistency parameters across multiple images alongside geometric positioning. The core of this approach lies in utilizing physical radiometric references to constrain the global energy optimization process, ensuring that the correction results are consistent not only statistically and visually but also at the physical radiometric level, thereby eliminating residual local color discrepancies.

The rest of this article is organized as follows. Section II systematically reviews the related work and analyzes existing limitations; Section III details the proposed radiometric and color balancing method based on remote sensing radiometric references; Section IV evaluates the performance of the proposed method quantitatively and qualitatively through rigorous experiments and comparisons with various state-of-the-art methods. Section V concludes this article and outlines directions for future research.

II. Related Work

Research on radiometric and color balancing for aerial imagery has evolved into diverse technical approaches. Based on their processing objects and core principles, these methodologies can be broadly categorized into two major classes: single-image nonuniformity correction methods and multi-image global color consistency correction methods. This section reviews and assesses representative works in these two directions, highlighting their contributions and identifying remaining limitations.

A. Single-Image Nonuniformity Color Correction Methods

Nonuniform illumination in single aerial images typically manifests as edge vignetting, local overexposure or underexposure, and local color casts within complex lighting environments. To address these issues, existing methods primarily achieve color uniformity through image decomposition, global/local statistical characteristic adjustment, or color mapping.

1) Statistical Transformation-Based Methods: Methods such as the histogram equalization algorithm [1] and the histogram matching algorithm [2] achieve color balance and enhance local contrast by stretching the grayscale histogram. Similarly, the Wallis algorithm [3], [4], [5] adjusts the mean and variance of the target image to match those of a reference image. The advantages of these methods lie in their computational simplicity and effectiveness in addressing general brightness nonuniformity. However, they are fundamentally data-driven statistical operations, detached from physical radiometric processes. In regions with low brightness or complex nonuniform illumination, global mapping may cause excessive enhancement or compression, leading to detail loss or halo artifacts, and often failing to avoid local color casts.

2) Decomposition-Based Methods: The Mask algorithm [5], [6], [7] achieves color uniformity by separating background and reflectance components via filtering. Homomorphic filtering algorithm [8] and Retinex-based algorithms [9], [10], [11], [12] decompose images into illumination and reflectance components, and suppress or remove the illumination component to extract uniform reflectance images. Although these methods incorporate physical intuition, existing models typically rely on simplified assumptions about illumination and reflectance, and the decomposition is inherently ill-posed—heavily dependent on priors and regularization terms. Consequently, under real complex scenarios, decomposition results often deviate from ground truth, leading to halos, blurred details, or color distortions in the corrected images.

3) Color Mapping-Based Methods: These methods achieve color correction by selecting a uniform reference image and establishing a mapping relationship between the target and reference images. Common mapping models include blockwise linear correction model [13], [14], probabilistic moving least-squares model [15], and orthogonal linear regression model [16]. Although these methods achieve good performance in certain scenarios, their mappings primarily target overall or blockwise color distributions. For issues such as nonuniform illumination and edge vignetting, a single mapping model is often insufficient for complete correction, potentially resulting in residual local color casts. Moreover, as these methods are based on statistical mapping rather than physical radiometric models, the corrected results may not necessarily conform to the true radiometric characteristics of ground objects.

4) Deep Learning-Based Methods: In recent years, deep learning has emerged as a robust paradigm for radiometric and color balancing by utilizing end-to-end networks to model complex nonlinear mappings. Early explorations predominantly relied on convolutional neural networks (CNNs) to directly regress color transformations [17], [18], which subsequently evolved into multitask architectures capable of concurrently handling various image enhancements [19]. To incorporate physical illumination priors, deep Retinex-Net [20] was introduced to decouple images into illumination and reflectance components, while recent advancements have tailored multichannel CNNs with residual learning specifically for UAV imagery [21]. Beyond direct regression, generative adversarial networks (GANs) and style transfer techniques have been widely adopted to mitigate radiometric

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disparities. Notable approaches include attention-guided color consistency GANs [22], unsupervised attention-fused CycleGANs [23], and WGAN-GP applications for enhancing visual fidelity [24]. For multitemporal applications, RS-NormGAN [25] has further advanced radiometric normalization through explicit cycle-consistency constraints. To explicitly preserve high-frequency details and structural integrity amidst these transformations, recent frameworks increasingly integrate semantic priors and advanced attention mechanisms. Methods range from local-to-global attention for semantic style transfer (DCC) [26] and semantic segmentation-enhanced color balance networks (SECBNets) [27], to cross-frequency attention with color contrast constraints [28]. Furthermore, cutting-edge representations such as neural radiance fields (NeRFs) are currently being explored to enable 3-D-aware color consistency correction in complex remote sensing scenarios [29]. While these state-of-the-art deep learning methods demonstrate superior performance in pixel-level color restoration and single-image visual enhancement, several critical challenges remain when applied to large-scale aerial mosaicking tasks. First, most current deep learning approaches focus on image-to-image transfer or local enhancement, lacking a global optimization mechanism that spans hundreds of images across multiple flight strips. This inevitably leads to residual color discrepancies and error accumulation at the boundaries of large-scale mosaics. Second, despite the introduction of semantic and attention modules, these data-driven models frequently operate as "black boxes" without explicit global physical radiometric constraints. This makes it difficult to guarantee high radiometric fidelity and uniform physical interpretability across complex, varying terrains. These unresolved issues emphasize the necessity of embedding reliable physical radiometric references into a global optimization framework, which is the exact motivation for the proposed radiometric aerotriangulation method.

In summary, while existing single-image nonuniformity color correction methods have made noticeable progress in enhancing visual consistency, their primary limitation lies in an over-reliance on image statistical properties or simplified/idealized physical models, failing to establish physical models that accurately describe real imaging radiometric distortions. In complex scenes, these methods may suffer from detail loss and local color casts. Although some deep learning-based methods have emerged, current approaches mostly focus on single-image enhancement or color transfer between image pairs, lacking systematic optimization designs for global color consistency across large-scale multi-image datasets.

B. Multi-Image Global Color Consistency Correction Methods

Multi-image global color consistency correction methods aim to eliminate color discrepancies across multiple images during mosaicking. Existing approaches are primarily categorized into methods based on overlapping region constraints and global energy optimization.

1) Methods Based on Overlapping Region Constraints: These methods utilize tie points or information from overlapping areas between images to establish local color transfer relationships, thereby achieving color transitions between adjacent images. Representative examples include blockwise Wallis algorithm [30], hybrid histogram matching (HHM) algorithm [31], conditional probability-based linear regression [32], weighted blockwise brightness compensation with adjacent-image brightness approximation [33], local estimated scatterplot smoothing (LOESS) radiometric correction for contiguous scenes (LORACCS) [34], and grouped histogram matching algorithm [35]. These methods are computationally efficient and capable of achieving smooth transitions in overlapping regions. However, their core limitation lies in the reliance solely on local overlap information. As correction information is propagated sequentially through adjacent images, severe error accumulation may occur, making it difficult to guarantee global optimal consistency. Furthermore, local linear or simple nonlinear models (e.g., LOESS) struggle to accurately fit complex physical radiometric discrepancies.

2) Multi-Image Global Energy Optimization Methods: These methods employ the prior knowledge that the same geographic location should share consistent color values. They minimize an energy function across the entire survey area to minimize global color differences. To maintain image quality, constraints such as gradient preservation are typically incorporated. Representative methodologies include multiview color consistency correction algorithm [36], global optimization with guided initial solutions [37], global color mapping models based on quadratic splines [38], [39], [40], block-adjustment methods [41], [42], and maximum a posteriori (MAP) estimation [43]. These approaches successfully circumvent error accumulation and substantially enhance global color consistency. Nevertheless, they primarily optimize the consistency of pixel value statistical distributions or visual continuity among images. By adopting a unified mapping model for an entire image, they often fail to effectively eliminate local color discrepancies. Although some studies have attempted to introduce local correction strategies—such as blockwise Gamma correction [44], uneven illumination removal based on bright channel prior [45], and blockwise linear correction [46], [47], [48]—to rectify residual color casts, limitations remain. Local correction models (e.g., Gamma correction and linear mapping models) inherently lack explicit physical radiometric significance. Moreover, they may introduce new issues such as "blocking artifacts," increased computational complexity, degradation of original spectral features, and loss of image details.

In general, radiometric and color balancing methods based on overlapping region constraints improve the overall color consistency of multiple images but are prone to error accumulation. Conversely, global energy optimization methods demonstrate significant effectiveness in overall consistency and represent the current mainstream direction. However, existing methods predominantly utilize data-driven statistical models as color mapping models, which tends to result in residual local color discrepancies and difficulties in ensuring physical consistency. There is currently a lack of a unified solution based on physical radiometric references that can simultaneously address detail preservation, accurate local color compensation, and global consistency optimization within a single framework.

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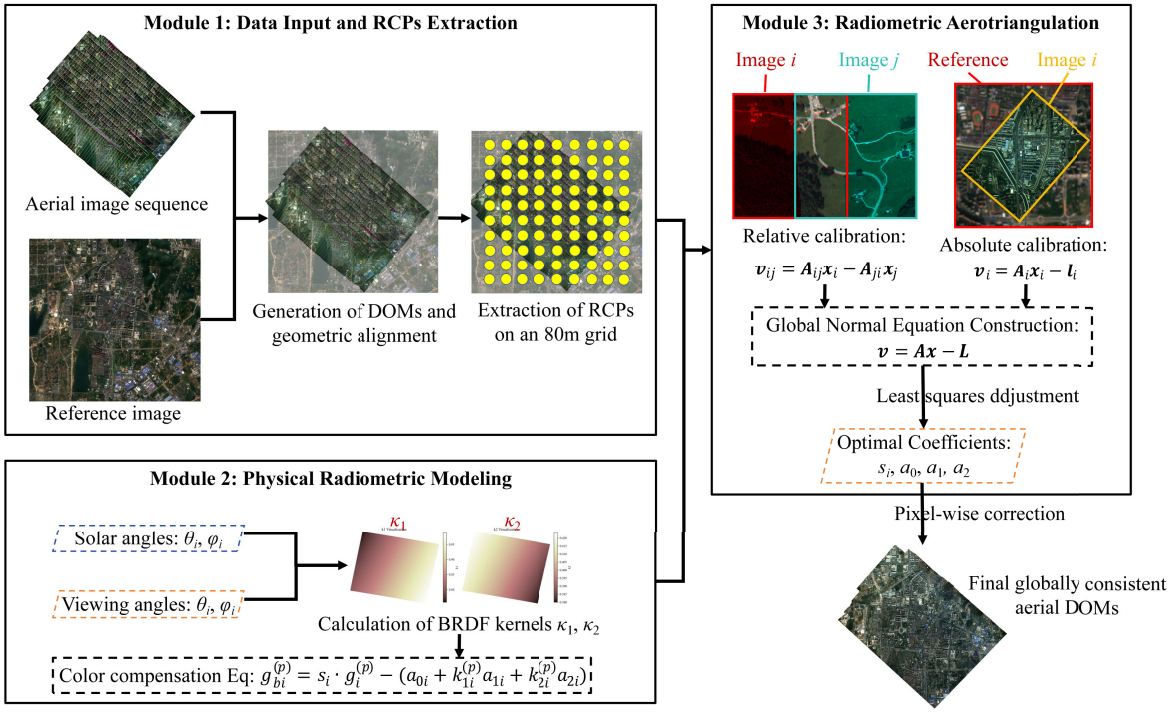


Fig. 1. Technical workflow of the proposed method. The framework consists of three main modules: 1) extraction of RCPs using an 80m geographic grid; 2) physical radiometric modeling using BRDF kernels derived from illumination and viewing geometries; and 3) global color consistency optimization via radiometric aerotriangulation, which jointly solves absolute and relative color constraints.

III. Methodology

To address the challenges of detail loss and local color casts associated with existing methods for aerial imagery nonuniformity color correction, this article proposes a radiometric and color balancing method based on remote sensing radiometric references. The proposed method initially acquires a remote sensing radiometric reference image and extracts tie points from all images to function as radiometric control points (RCPs). Subsequently, a radiometric equation for nonuniformity color compensation is established. Based on the condition that the color values of tie points should be consistent across different images, color consistency error equations based on radiometric aerotriangulation are formulated. Next, normal equations are constructed based on the least-squares principle to solve for the color correction coefficients of each image. Finally, pixel-wise correction is performed on each image using the calculated coefficients, and all images are mosaicked to generate the final output product. The workflow of the proposed method is illustrated in Fig. 1.

It is important to clarify the terminology used in this article. Fundamentally, the proposed method constitutes a global least-squares adjustment bounded by physical radiometric constraints. We adopt the term "radiometric aerotriangulation" as an analogy to traditional photogrammetric geometric block adjustment. In traditional geometric aerotriangulation, ground control points (GCPs) and tie points are utilized to jointly optimize camera poses and 3-D coordinates. Analogously, in our radiometric framework, RCPs act as absolute radiometric constraints, while overlapping tie points serve as relative radiometric constraints. By constructing a joint observation equation system, we globally optimize the radiometric compensation coefficients for the entire image block. Therefore, while fundamentally different from geometric optimization, our method inherits the robust global error-propagation control inherent in the mathematical architecture of classical aerotriangulation.

A. Extraction of RCPs

Prior to extracting RCPs, it is necessary to acquire a remote sensing radiometric reference image. Currently, radiometric references primarily stem from two sources: ground-measured data and radiometric products with known reflectance. While the former is difficult to obtain, the latter can be selected from satellite image products acquired synchronously or within a time window of about one week under similar weather and illumination conditions (e.g., Landsat Level-2 products and Sentinel-2 Level-2A products). These standard radiometric products undergo rigorous atmospheric correction and radiometric calibration, achieving a relative radiometric uncertainty of typically less than 3%–5% [49], [50]. Furthermore, a single scene of these satellite images covers a vast geographic area, which is generally significantly larger than the coverage of a typical aerial photogrammetry block. Consequently, for most aerial photogrammetry tasks, the radiometric reference originates from a single, internally consistent satellite scene. This inherent consistency ensures that the reference itself does not introduce spatial segmentation artifacts or "block effects" into the mosaicking process of aerial imagery.

Existing radiometric and color balancing methods are mainly optimized for and applied to DOMs. Accordingly, this

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article proposes a one-time extraction strategy for uniformly distributed RCPs based on an orthorectified grid. The process comprises the following steps.

1) *Generation of DOMs From Original Images:* This step can be accomplished using existing software, such as the DPGrid software developed by the team of Academician Zhang Zuxun at Wuhan University.
2) *Alignment of Generated DOMs With the Reference Image:* The reference image is a satellite reflectance product that has undergone geometric correction and is inherently orthorectified. Therefore, by strictly aligning all images within a unified geographic coordinate system, a one-to-one correspondence among pixels, geographic locations, and image values can be established.
3) *Construction of a Ground Grid and Extraction of RCPs:* A unified geographic grid (e.g., a grid spacing of 80 m) is generated within the spatial extent of the DOMs. For each grid point, the corresponding pixel values are extracted from both the reference image and each aerial orthophoto. For aerial imagery, as individual pixel values are susceptible to noise and texture variations, mean values from a local neighborhood are calculated, thereby obtaining a more representative radiometric measurement. A grid point covered by multiple images is considered a tie point across those images and is thus treated as an RCP.
4) *Filtration of RCPs:* Points with a pixel value of 0 have no physical radiometric meaning and may adversely affect fitting results during color consistency correction. Therefore, any control point that has a zero value in any band of any image is removed.

Through the above approach, a large number of stable and uniformly distributed RCPs can be extracted simultaneously from multiple aerial images. Given the minimal temporal difference between the acquisition of aerial and reference images, coupled with consistent weather conditions, ground object changes are negligible; thus, further removal of land-cover change points is unnecessary.

Furthermore, the total quantity of RCPs is not fixed; rather, it is dynamically determined by the spatial extent of the survey area and the predefined geographic grid interval (set to 80 m in this study). The choice of grid density dictates the tradeoff between computational efficiency and local color correction accuracy. A sparser grid (e.g., 160 m) reduces the number of observation equations, accelerating the least-squares adjustment, but risks underfitting local illumination variations. Conversely, an overly dense grid (e.g., 40 m) introduces redundant observations that exponentially increase the dimensions of the normal equations without yielding proportional accuracy gains. Empirical evaluations indicate that an 80-m sampling interval achieves an optimal balance, providing sufficient physical constraints for localized color compensation while ensuring efficient processing for large-scale datasets.

B. Physical Radiometric Model for Nonuniformity Color Compensation

To address the issues of detail loss and local color casts in single-image nonuniformity correction caused by the absence of accurate physical radiometric models, this article proposes a physical radiometric model based on the bidirectional reflectance distribution function (BRDF) for color compensation. The model explicitly incorporates the influence of imaging geometric factors (e.g., solar incidence angle, surface reflection angle, and camera viewing angle) on illumination distribution. It characterizes the formation mechanism of brightness and color variations in imagery from a physical perspective, thereby successfully eliminating intraimage brightness nonuniformity and color casts while preserving texture details.

For aerial imagery acquired during the same period, the primary causes of color nonuniformity within or across images include illumination conditions, terrain undulation, and camera attitude. In most common scenarios, color discrepancies induced by terrain undulation are relatively minor. Consequently, this study focuses on correcting color inconsistencies caused by variations in illumination conditions and camera attitude. Given that color information in imagery is significantly influenced by illumination direction, it is imperative to account for imaging geometric factors—including the solar incidence angle, surface reflection angle, and camera viewing angle—during nonuniform color correction. Such geometric radiometric modeling based on imaging angles is typically implemented via BRDF to effectively compensate for brightness and color variances arising from changes in illumination and viewing angles. For any point p on image i ($i = 1, 2, \ldots, N$, where N denotes the number of images to be corrected), the radiometric equation used for color compensation is formulated as

$$g_{bi}^{(p)} = s_i \cdot g_i^{(p)} - \left(a_{0i} + k_{1i}^{(p)} a_{1i} + k_{2i}^{(p)} a_{2i}\right) \tag{1}$$

where $g_{bi}^{(p)}$ represents the corrected color value; $g_i^{(p)}$ denotes the original color value; s_i is a scale factor, which can be solved through the error equation adjustment; $k_{1i}^{(p)}$ and $k_{2i}^{(p)}$ are BRDF kernels related to the solar incidence angle and camera viewing angle, primarily addressing the issue of nonuniform illuminance distribution within the camera's field of view; and a_{0i}, a_{1i}, and a_{2i} are coefficients of the BRDF kernels. The BRDF kernels $k_{1i}^{(p)}$ and $k_{2i}^{(p)}$ are computed based on the linear semi-empirical RossThick-LiSparse (RTLSR) model [51], [52] as follows:

$$\begin{cases} k_{1i}^{(p)} = \dfrac{\left(\frac{\pi}{2} - \xi_i^{(p)}\right)\cos\xi_i^{(p)} + \sin\xi_i^{(p)}}{\cos\theta_{si}^{(p)} + \cos\theta_{vi}^{(p)}} - \dfrac{\pi}{4} \\ k_{2i}^{(p)} = O_i^{(p)} - \sec\theta_{si}^{\prime(p)} - \sec\theta_{vi}^{\prime(p)} + \dfrac{1}{2}\left(1 + \cos\xi_i^{\prime(p)}\right)\sec\theta_{vi}^{\prime(p)} \end{cases} \tag{2}$$

where $k_{1i}^{(p)}$ is the Ross–Thick volume scattering kernel and $k_{2i}^{(p)}$ is the Li-Sparse geometric-optical scattering kernel. The intermediate parameters in (2) are calculated as follows.

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1) Phase Angle: The phase angle $\xi_i^{(p)}$ represents the 3-D angle between the solar incidence direction and the camera viewing direction

$$\cos\xi_i^{(p)} = \cos\theta_{si}^{(p)}\cos\theta_{vi}^{(p)} + \sin\theta_{si}^{(p)}\sin\theta_{vi}^{(p)}\cos\left(\phi_{vi}^{(p)} - \phi_{si}^{(p)}\right) \tag{3}$$

where $\theta_{si}^{(p)}$ and $\theta_{vi}^{(p)}$ are the solar and viewing zenith angles, respectively, and $\phi_{si}^{(p)}$ and $\phi_{vi}^{(p)}$ are the solar and viewing azimuth angles.

2) Modified Zenith Angles and Modified Phase Angle: To account for the geometric effects of the ellipsoidal crown shape, the zenith angles are modified to θ' using the crown shape parameter b/r (the ratio of vertical crown radius to horizontal crown radius)

$$\begin{cases} \theta_{si}'^{(p)} = \arctan\left(\dfrac{b}{r}\tan\theta_{si}^{(p)}\right) \\ \theta_{vi}'^{(p)} = \arctan\left(\dfrac{b}{r}\tan\theta_{vi}^{(p)}\right). \end{cases} \tag{4}$$

The modified phase angle $\xi_i'^{(p)}$ for the geometric kernel is then computed as

$$\cos\xi_i'^{(p)} = \cos\theta_{si}'^{(p)}\cos\theta_{vi}'^{(p)} + \sin\theta_{si}'^{(p)}\sin\theta_{vi}'^{(p)}\cos\left(\phi_{vi}^{(p)} - \phi_{si}^{(p)}\right). \tag{5}$$

3) Overlap Area: The overlap area $O_i^{(p)}$ between the shadow and the illuminated area is defined as

$$O_i^{(p)} = \frac{1}{\pi}\left(t_i^{(p)} - \sin t_i^{(p)}\cos t_i^{(p)}\right)\left(\sec\theta_{si}'^{(p)} + \sec\theta_{vi}'^{(p)}\right) \tag{6}$$

in which $\cos t_i^{(p)}$ is determined by

$$\cos t_i^{(p)} = \frac{h}{b}\cdot\left(\sec\theta_{si}'^{(p)} + \sec\theta_{vi}'^{(p)}\right)^{-1} \times\left[D_i^{(p)2} + \left(\tan\theta_{si}'^{(p)}\tan\theta_{vi}'^{(p)}\sin\left(\phi_{vi}^{(p)} - \phi_{si}^{(p)}\right)\right)^2\right]^{\frac{1}{2}} \tag{7}$$

where $D_i^{(p)}$ is the horizontal distance between the centers of the crown and its shadow and h/b is the ratio of crown center height to vertical crown radius. In this study, the canopy shape parameters are set to $b/r = 1.6$ and $h/b = 2.3$ as empirical defaults for aerial scenarios. $D_i^{(p)}$ is computed as follows:

$$D_i^{(p)} = \left[\tan^2\theta_{si}'^{(p)} + \tan^2\theta_{vi}'^{(p)} - 2\tan\theta_{si}'^{(p)}\tan\theta_{vi}'^{(p)}\cos\left(\phi_{vi}^{(p)} - \phi_{si}^{(p)}\right)\right]^{\frac{1}{2}}. \tag{8}$$

Specifically, the scaling factor s_i functions as a linear radiometric normalizer to bridge the inherent dynamic range gap between disparate sensors. For instance, the reference Sentinel-2 Level-2A imagery typically features surface reflectance digital numbers (DNs) up to 10 000, whereas UAV imagery typically operates within a 12-bit (0–4095) range. The scaling factor s_i effectively aligns the UAV radiometric scale with the global reference prior to BRDF compensation.

It should be noted that the BRDF kernel functions $k_{1i}^{(p)}$ and $k_{2i}^{(p)}$ in (2) are defined based on the classic linear semi-empirical BRDF modeling framework, originally pioneered by Roujean et al. [53]. In our implementation, we adopt the widely recognized RTLSR model [51], which forms the theoretical foundation of the standard algorithm for MODIS global BRDF products. Here, $k_{1i}^{(p)}$ represents the Ross–Thick volume scattering kernel, which models radiative transfer within dense canopies, while $k_{2i}^{(p)}$ represents the original Li-Sparse geometric-optical scattering kernel, which accounts for the mutual shadowing effects of surface structures (such as trees and buildings) driven by illumination and viewing geometries. By utilizing these physical kernels, our model effectively approximates complex surface reflectance behaviors while maintaining the linear mathematical formulation, which is crucial for the highly efficient computational optimization required in global radiometric aerotriangulation.

C. Color Consistency Correction Model Based on Radiometric Aerotriangulation

To address the issues of lacking physical radiometric reference constraints in multi-image global color consistency correction and the difficulty in completely eliminating local color discrepancies, this article proposes a "Radiometric Aerotriangulation" model inspired by geometric aerotriangulation. This method integrates RCPs and their corresponding BRDF kernel parameters into color consistency error equations. By jointly optimizing absolute and relative color corrections, the model ensures global color consistency while maintaining physical radiometric fidelity, successfully eliminating local residuals and suppressing error accumulation.

Absolute color correction focuses on the calibration between the target image and the reference image. Utilizing the least-squares principle, it aims to minimize the difference between the corrected image colors and the reference image. For each pair of tie points between image i and the reference image, the BRDF kernels are calculated and substituted into (1) to derive the absolute color correction error equations for image i

$$\begin{cases} v_i^{(1)} = a_{0i} + k_{1i}^{(1)}a_{1i} + k_{2i}^{(1)}a_{2i} - g_i^{(1)}\cdot s_i + g_{bi}^{(1)} \\ v_i^{(2)} = a_{0i} + k_{1i}^{(2)}a_{1i} + k_{2i}^{(2)}a_{2i} - g_i^{(2)}\cdot s_i + g_{bi}^{(2)} \\ \quad\vdots \\ v_i^{(m)} = a_{0i} + k_{1i}^{(m)}a_{1i} + k_{2i}^{(m)}a_{2i} - g_i^{(m)}\cdot s_i + g_{bi}^{(m)} \end{cases} \tag{9}$$

where $v_i^{(u)}(u = 1, 2, \ldots, m)$ are observation residuals and m is the number of tie-point pairs between image i and the reference image. Equation (9) can be simplified into the following matrix form:

$$\boldsymbol{v}_i = \boldsymbol{A}_i\boldsymbol{x}_i - \boldsymbol{l}_i \tag{10}$$

where $\boldsymbol{v}_i$ is the residual vector, $\boldsymbol{A}_i$ is the coefficient matrix, $\boldsymbol{x}_i$ is the vector of unknowns, and $\boldsymbol{l}_i$ is the observation vector.

Relative color correction addresses the calibration between different target images. It employs the least-squares principle to maximize color consistency in overlapping regions, thereby achieving overall color uniformity. For any tie point q shared

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by images i and $j(j = 1, 2, \ldots, N; j \neq i)$, the corrected color values must satisfy (1) in both images

$$s_i \cdot g_i^{(q)} - \left(a_{0i} + k_{1i}^{(q)} a_{1i} + k_{2i}^{(q)} a_{2i}\right) = s_j \cdot g_j^{(q)} - \left(a_{0j} + k_{1j}^{(q)} a_{1j} + k_{2j}^{(q)} a_{2j}\right). \tag{11}$$

The BRDF kernels for each pair of tie points are calculated and substituted into (11) to derive the error equations for relative color calibration between images i and j

$$\begin{cases} v_{ij}^{(1)} = a_{0i} + k_{1ij}^{(1)} a_{1i} + k_{2ij}^{(1)} a_{2i} - g_{ij}^{(1)} \cdot s_i \\ \qquad\qquad - a_{0j} - k_{1ji}^{(1)} a_{1j} - k_{2ji}^{(1)} a_{2j} + g_{ji}^{(1)} \cdot s_j \\ v_{ij}^{(2)} = a_{0i} + k_{1ij}^{(2)} a_{1i} + k_{2ij}^{(2)} a_{2i} - g_{ij}^{(2)} \cdot s_i \\ \qquad\qquad - a_{0j} - k_{1ji}^{(2)} a_{1j} - k_{2ji}^{(2)} a_{2j} + g_{ji}^{(2)} \cdot s_j \\ \qquad\qquad \vdots \\ v_{ij}^{(n)} = a_{0i} + k_{1ij}^{(n)} a_{1i} + k_{2ij}^{(n)} a_{2i} - g_{ij}^{(n)} \cdot s_i \\ \qquad\qquad - a_{0j} - k_{1ji}^{(n)} a_{1j} - k_{2ji}^{(n)} a_{2j} + g_{ji}^{(n)} \cdot s_j \end{cases} \tag{12}$$

where $v_{ij}^{(w)}(w = 1, 2, \ldots, n)$ are observation residuals; n is the number of tie-point pairs between images i and j; $k_{1ij}^{(w)}, k_{2ij}^{(w)}$ and $k_{1ji}^{(w)}, k_{2ji}^{(w)}$ are BRDF kernels for tie point w in images i and j, respectively; $g_{ij}^{(w)}$ and $g_{ji}^{(w)}$ are the color values of w in images i and j; a_{0j}, a_{1j}, and a_{2j} are coefficients for image j; and s_j is a scale factor for image j. Equation (12) can be simplified into the following matrix form:

$$\boldsymbol{v}_{ij} = \boldsymbol{A}_{ij}\boldsymbol{x}_i - \boldsymbol{A}_{ji}\boldsymbol{x}_j \tag{13}$$

where $\boldsymbol{v}_{ij}$ is the residual vector, $\boldsymbol{A}_{ij}$ and $\boldsymbol{A}_{ji}$ are the coefficient matrices, and $\boldsymbol{x}_j$ is the vector of unknowns for image j.

Absolute color correction equations are constructed for each image, and relative correction equations are added for each image pair whose overlap exceeds a predefined threshold (recommended 10%). All error equations are combined to form a unified global error equation

$$\boldsymbol{v} = \boldsymbol{A}\boldsymbol{x} - \boldsymbol{L} \tag{14}$$

where $\boldsymbol{v}$ is the total residual vector; $\boldsymbol{A}$ is the coefficient matrix for all unknowns, which is a banded sparse matrix (to improve solution efficiency, adjacent-image pairs are ordered sequentially during construction); $\boldsymbol{x}$ is the vector of all unknowns; and $\boldsymbol{L}$ is the observation vector. According to the principle of least-squares indirect adjustment, the normal equation can be formulated as

$$\boldsymbol{A}^T\boldsymbol{P}\boldsymbol{A}\boldsymbol{x} = \boldsymbol{A}^T\boldsymbol{P}\boldsymbol{L} \tag{15}$$

where $\boldsymbol{P}$ is the weight matrix of the observations, reflecting measurement precision. Generally, color observations for all image points are considered to have equal precision, making $\boldsymbol{P}$ an identity matrix. The solution to the equation can be obtained as

$$\boldsymbol{x} = \left(\boldsymbol{A}^T\boldsymbol{A}\right)^{-1}\boldsymbol{A}^T\boldsymbol{L}. \tag{16}$$

Finally, the calculated color correction coefficients are applied to perform pixel-wise correction on each image. All corrected images are then mosaicked to produce a final output that exhibits both global and local consistency.

It should be noted that the proposed method estimates radiometrically corrected reflectance values rather than display-ready RGB intensities. To generate true-color images for visualization and evaluation, the corrected reflectance values are mapped to the RGB color space using the standard band-to-channel assignment and linear stretching commonly adopted in remote sensing visualization. Since this mapping is monotonic and applied consistently across all images, it does not alter the relative radiometric relationships or color consistency achieved by the proposed correction.

IV. Experiment Results and Discussion

To validate the effectiveness of the proposed method in improving the brightness and color consistency of aerial imagery while preserving ground radiometric features and texture details, experiments were conducted using three typical aerial photography datasets. The comparative methods include traditional radiometric and color balancing methods (Wallis [5] and Retinex [9]), statistical methods based on overlapping region constraints (HHM [31] and LORACCS [34]), and advanced deep learning-based approaches (DCC [26], C4RSD [28], and RS-NormGAN [25]). The comparison encompasses intraimage color balance and brightness consistency for single images, global consistency of multi-image mosaicking results, and local continuity and color consistency in seam regions. The evaluation metrics include color distance (CD), mean squared error (mse), peak signal-to-noise ratio (PSNR), and DeltaE [54].

A. Datasets and Evaluation Metrics

The radiometric reference used in this study is based on Sentinel-2 Level-2A surface reflectance products, which are atmospherically corrected and radiometrically calibrated standard products. The proposed color consistency correction method was evaluated on three datasets, whose detailed specifications are listed in Table I. Each image in the DMC dataset was obtained by stitching multiple subimages captured by high-resolution large-format cameras. Specifically, the DMC dataset was collected by the Leica DMC-4 camera, while datasets DMZ1 and DMZ2 were acquired using the high-resolution large-format camera (referred to as GFDMZ) developed by the No. 508 Institute of the 5th Academy of the China Aerospace Science and Technology Corporation. High-resolution large-format cameras possess large-footprint imaging capabilities, high spatial resolution, high geometric precision, and superior radiometric performance, enabling the acquisition of aerial imagery with wider coverage and clearer details in a single flight. While preserving the spectral and texture features of ground objects, such cameras substantially enhance data-acquisition efficiency and positioning accuracy, making them particularly suitable for large-scale high-precision mapping and 3-D modeling.

The images from the three datasets are shown in Fig. 2. The DMC dataset exhibits relatively uniform internal color distribution and good color consistency across multiple images, indicating high overall quality. In contrast, the DMZ1 dataset displays obvious color casts and intraimage nonuniformity, resulting in poor interimage color consistency. The

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TABLE I
PARAMETERS OF EXPERIMENTAL DATA

Dataset	Number	Width	Height	Platform	Spectral channel	Resolution (m)
DMC	1096	31520	13440	UAV	RGB+NIR	0.05
DMZ1	562	8424	6032	UAV	RGB+NIR	0.20
DMZ2	286	7920	6004	UAV	RGB+NIR	0.20

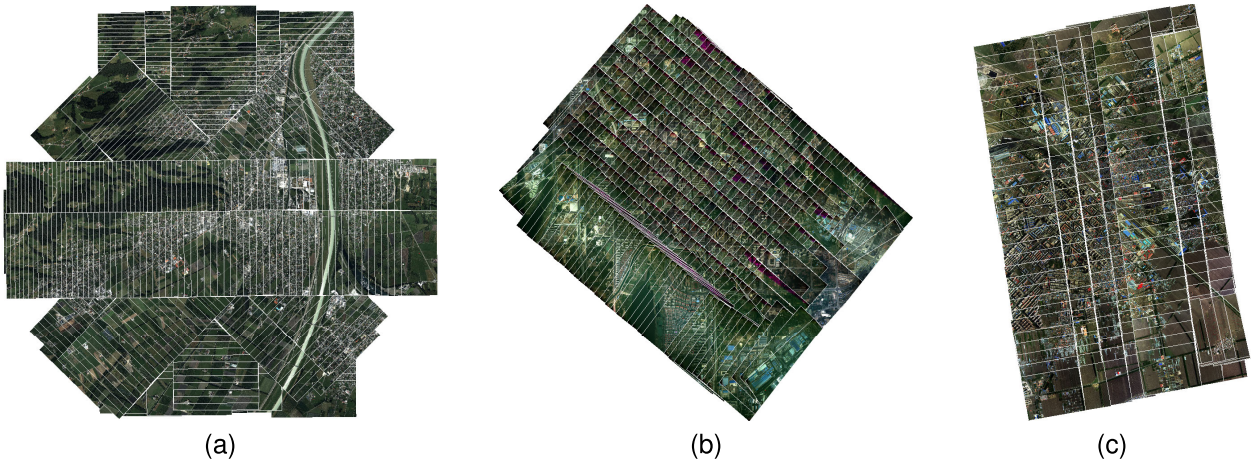


Fig. 2. Images of the three datasets. (a) DMC. (b) DMZ1. (c) DMZ2.

DMZ2 dataset is characterized by prominent local brightness nonuniformity and is particularly prone to significant color discrepancies at flight-strip intersections. These three datasets present varying degrees of uneven illumination, color deviation, and imaging stability differences, providing diverse experimental conditions to verify the applicability and robustness of the proposed method.

Furthermore, to quantitatively evaluate the performance of the color correction methods, this study selected CD, mse, PSNR, and DeltaE. Among these, CD, mse, and PSNR are employed to measure global color consistency in the overlapping regions of multiple images, while DeltaE is used to evaluate intraimage color balancing. The definitions of these metrics are as follows.

1) *CD:* It evaluates color consistency by calculating the histogram differences between overlapping regions. A smaller CD indicates better consistency. The calculation formula for CD is

$$\mathrm{CD} = \sum_{I_i \cap I_j \neq 0} w_{ij} \frac{\Delta H\left(I_{ij}, I_{ji}\right)}{N_b} \tag{17}$$

where I_i and I_j are two corrected images with overlapping regions; I_{ij} represents the region of I_i overlapping I_j, and similarly I_{ji} for I_j overlapping I_i; w_{ij} is a weight proportional to the overlapping area, with $\sum w_{ij} = 1$; $\Delta H(I_{ij}, I_{ji})$ represents the histogram distance between the two overlapping regions; and N_b denotes the number of histogram bins.

2) *MSE and PSNR:* MSE evaluates color consistency by measuring the pixel-level deviations between overlapping regions, with a smaller mse indicating better color consistency. The calculation formula for mse is

$$\mathrm{mse} = \frac{1}{M \times N} \sum_{m=1}^{M} \sum_{n=1}^{N} \left(I_{ij}(m,n) - I_{ji}(m,n)\right)^2 \tag{18}$$

where M and N denote the number of rows and columns in the overlapping region. The PSNR is calculated as follows:

$$\mathrm{PSNR} = 10 \times \log_{10} \frac{\mathrm{MAX}^2}{\mathrm{mse}} \tag{19}$$

where MAX represents the maximum possible pixel value. A higher PSNR indicates better consistency.

3) *DeltaE:* It (CIE2000 standard) [54] is currently one of the most accurate color difference measurement methods. A smaller DeltaE indicates better color uniformity. Due to its complexity, only the general form is provided

$$\Delta E = \left[\left(\frac{\Delta L'}{k_L S_L}\right)^2 + \left(\frac{\Delta C'}{k_C S_C}\right)^2 + \left(\frac{\Delta H'}{k_H S_H}\right)^2 + R_T \left(\frac{\Delta C'}{k_C S_C}\right)\left(\frac{\Delta H'}{k_H S_H}\right) \right]^{\frac{1}{2}} \tag{20}$$

where $\Delta L', \Delta C'$, and $\Delta H'$ represent difference in lightness, chroma, and hue; S_L, S_C, and S_H are weighting functions; k_L, k_C, and k_H are weighting coefficients; and R_T is the rotation term.

B. Qualitative Evaluation

The qualitative evaluation is conducted through visual inspection, focusing on two aspects: the intraimage color uniformity and the color consistency across seamlines in mosaicking results. The proposed method is compared with

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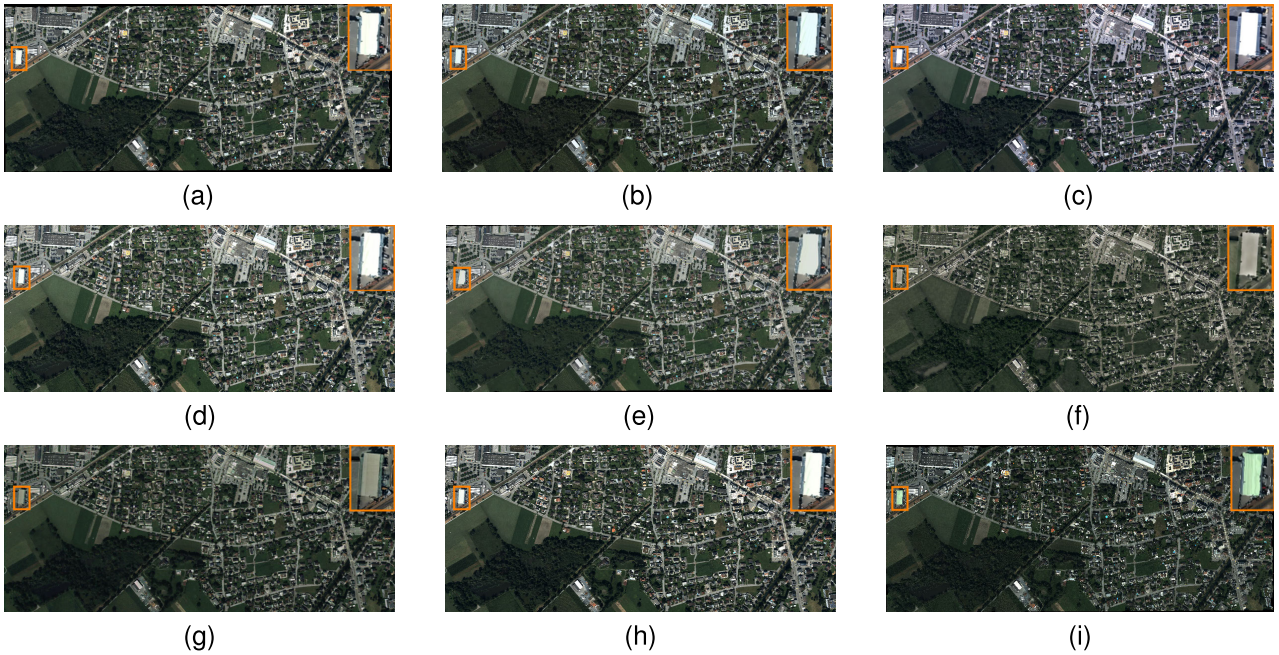


Fig. 3. Single-image color correction results for each method on the DMC dataset. (a) Original. (b) Wallis. (c) Retinex. (d) HHM. (e) LORACCS. (f) DCC. (g) C4RSD. (h) RS-NormGAN. (i) Ours.

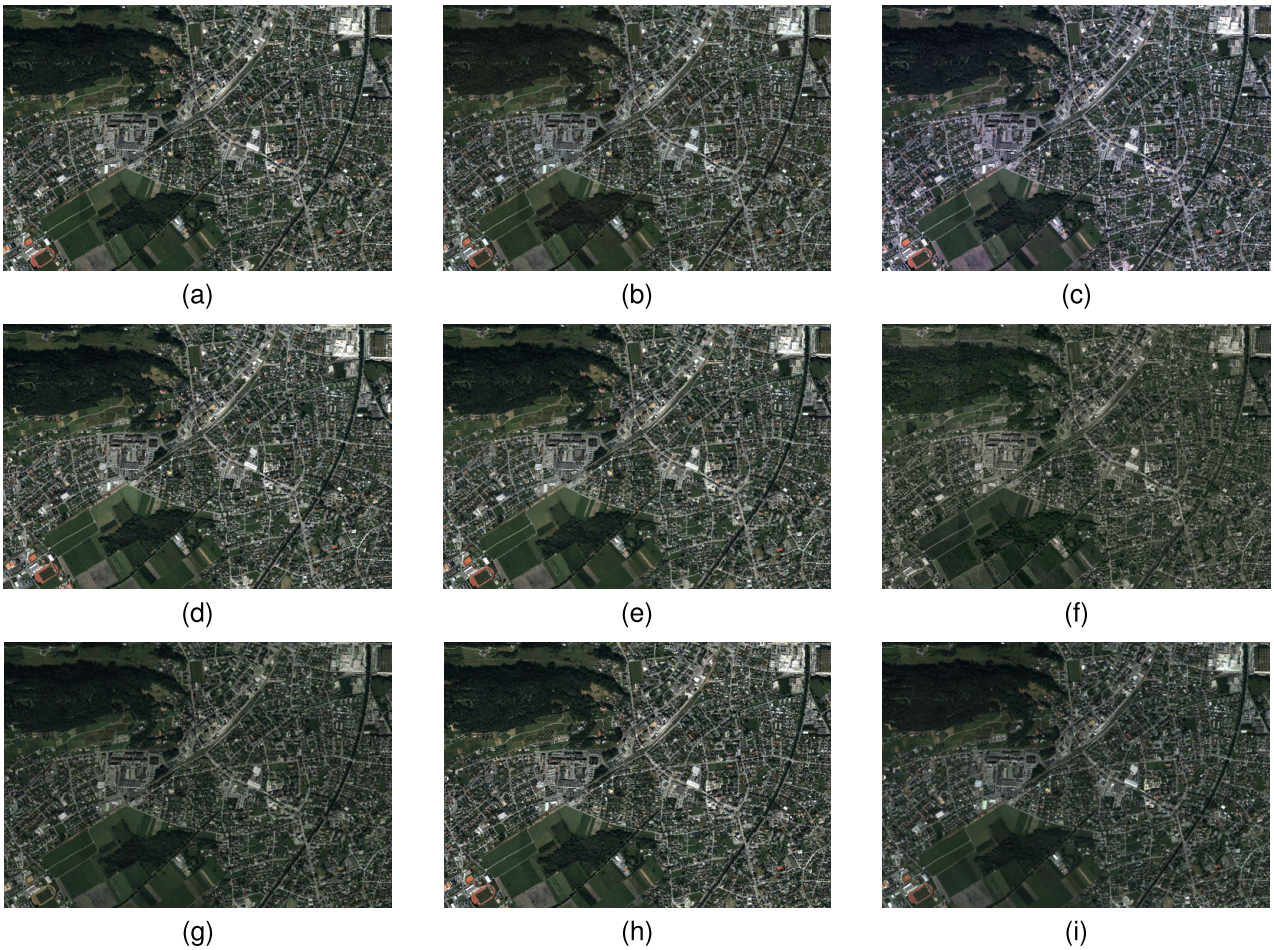


Fig. 4. Mosaicking results of selected images for each method on the DMC dataset. (a) Original. (b) Wallis. (c) Retinex. (d) HHM. (e) LORACCS. (f) DCC. (g) C4RSD. (h) RS-NormGAN. (i) Ours.

the Wallis algorithm, Retinex algorithm, HHM, LORACCS, DCC, C4RSD, and RS-NormGAN. The qualitative results are presented in Figs. 3–11. Specifically, Figs. 3, 6, and 9 present the color correction results of single images; Figs. 4, 7, and

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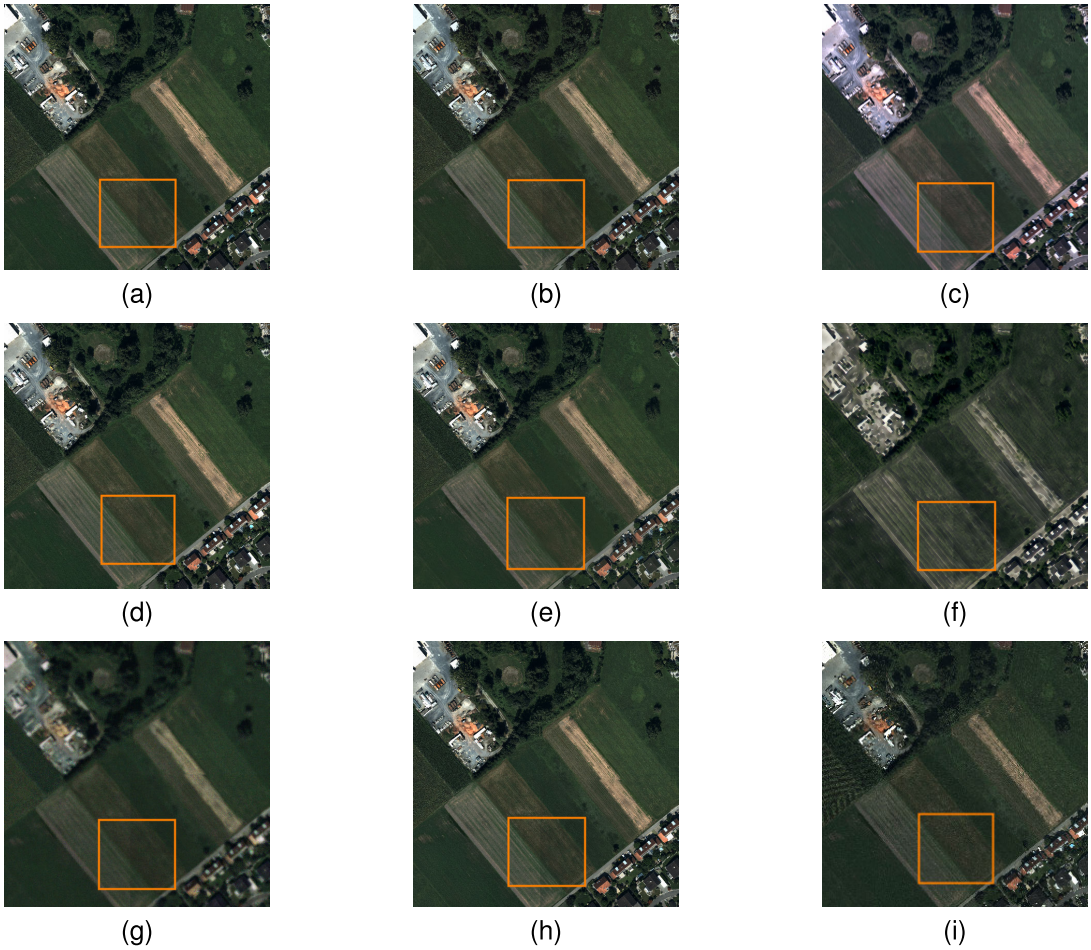


Fig. 5. Local details of mosaicking results for each method on the DMC dataset. (a) Original. (b) Wallis. (c) Retinex. (d) HHM. (e) LORACCS. (f) DCC. (g) C4RSD. (h) RS-NormGAN. (i) Ours.

10 display the overall mosaicking results; and Figs. 5, 8, and 11 illustrate local details of the mosaicking results. All images were individually corrected and subsequently mosaicked using DPGrid, a software developed by the team of Academician Zhang Zuxun at the School of Remote Sensing and Information Engineering, Wuhan University.

The DMC dataset exhibits relatively high overall quality with comparatively uniform color and brightness distribution within single images. Consequently, the differences in correction effects among the various methods on this dataset are relatively minor. As observed in Fig. 3, all methods achieved a high degree of color balance. Notably, the results obtained by Wallis, HHM, LORACCS, RS-NormGAN, and the proposed method maintain high visual consistency with the original image. The Retinex algorithm demonstrates a pronounced color enhancement effect, rendering the overall image more vivid. Conversely, DCC aims to resolve radiometric variations through neural semantic style transfer. However, due to the black-box nature of data-driven feature mappings, it inherently tends to introduce severe color casts over unmodeled complex terrestrial structures—for instance, DCC leads to noticeable color shifts and saturation distortions in building areas. Similarly, C4RSD exhibits a certain degree of color distortion over artificial materials such as buildings, as highlighted by the orange bounding boxes in Fig. 3, where the top-right inset of each subfigure presents a magnified view of the corresponding region for closer inspection. In contrast, by incorporating remote sensing radiometric references, the proposed method effectively circumvents color distortion on highly reflective architectural structures, successfully restoring their intrinsic physical colors. Since the DMC dataset contains 1096 images, the data volume is large. To provide a clear and representative display of overall mosaicking performance, 161 images were selected for demonstration, as shown in Fig. 4. Due to the inherent superior interimage color consistency of the DMC dataset, all methods achieve stable and coherent global mosaicking performance, without noticeable color jumps across images. Further observation of the zoomed-in detail in Fig. 5 reveals that HHM, LORACCS, and the proposed method demonstrate the best performance in brightness and color transitions at mosaicking boundaries, maintaining natural seamline connections. By comparison, the other methods still exhibit slight residual color discrepancies in local regions, as indicated by the orange bounding boxes in the figure. For the DMC dataset, where the original images already exhibit relatively uniform illumination and good interimage consistency, the advantage of the proposed method lies not in aggressive correction but in avoiding overcorrection. Owing to the physical constraints imposed by the radiometric model, unnecessary color stretching is suppressed, which helps

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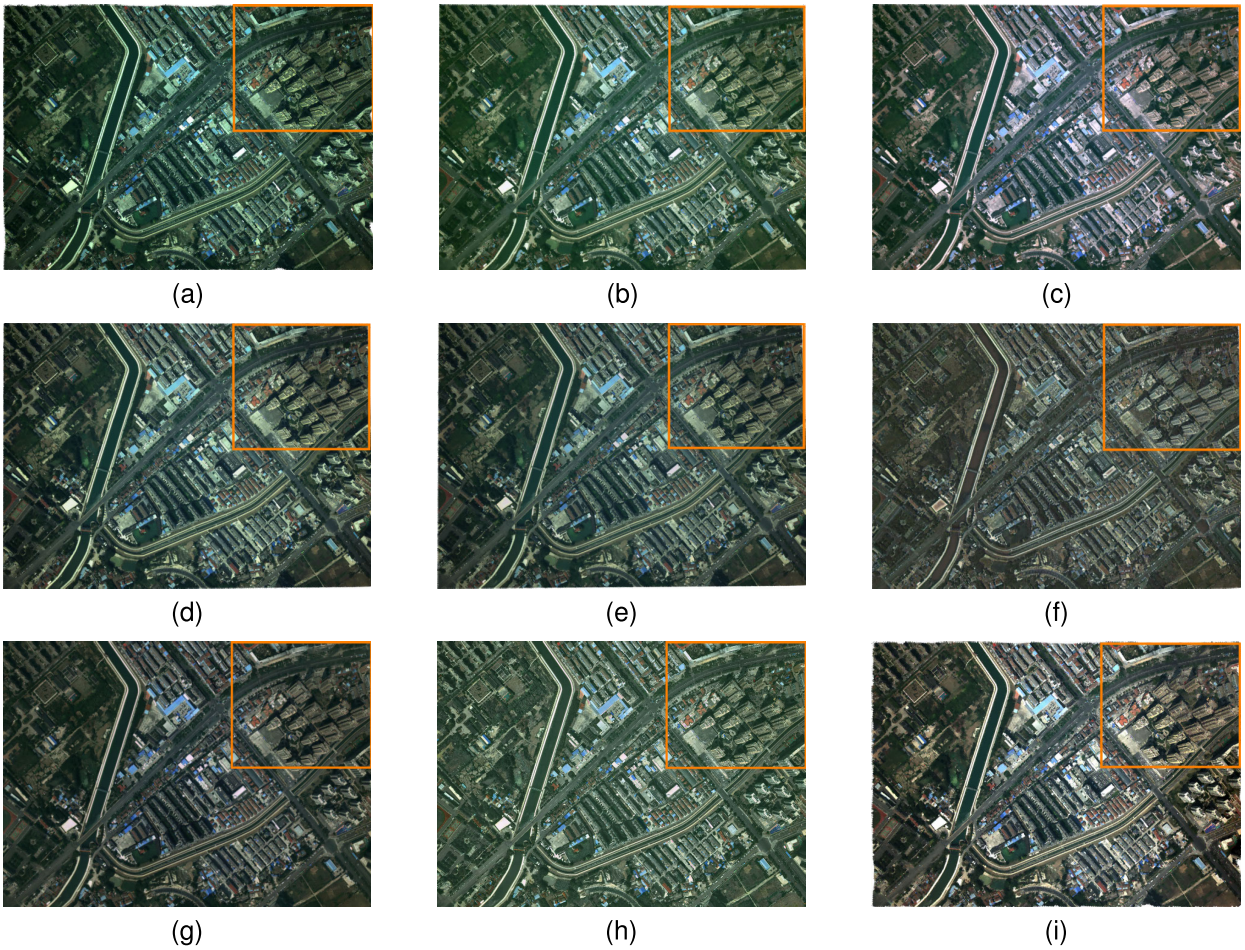


Fig. 6. Single-image color correction results for each method on the DMZ1 dataset. (a) Original. (b) Wallis. (c) Retinex. (d) HHM. (e) LORACCS. (f) DCC. (g) C4RSD. (h) RS-NormGAN. (i) Ours.

preserve the original radiometric characteristics and prevents artificial color enhancement.

The DMZ1 dataset exhibits significant color shifts and uneven illumination, serving as an effective benchmark for evaluating the robustness of various methods under complex radiometric conditions. As shown in Fig. 6, the Wallis algorithm, DCC, C4RSD, and RS-NormGAN alleviate local brightness nonuniformity to a certain extent. However, the Wallis algorithm fails to achieve color balance, while DCC exacerbates color distortion issues. The Retinex, HHM, and LORACCS algorithms improve color casts and yield visually superior images, yet the degree of improvement is limited, and they fail to effectively eliminate local brightness nonuniformity. This intraimage nonuniformity is explicitly evidenced by the distinct brightness discrepancy between the upper right and lower left regions within the orange bounding boxes in Fig. 6. In contrast, the proposed method not only significantly corrects color casts but also simultaneously resolves local brightness nonuniformity while preserving surface textures and fine details. Furthermore, Figs. 7 and 8 reveal that the Wallis, Retinex, C4RSD, and RS-NormGAN algorithms still exhibit noticeable color discrepancies in seam regions, failing to achieve smooth transitions between adjacent images. This boundary discontinuity is particularly prominent within the orange bounding boxes in Fig. 8, where sharp artificial seams remain unresolved. The primary bottleneck for C4RSD and RS-NormGAN in achieving seamless mosaics stems from their foundational architectural constraints. Their primary objective is strictly confined to single-image radiance recovery or independent image-level normalization. Consequently, they fundamentally lack interimage radiometric consistency constraints and joint global optimization mechanisms across adjacent overlapping flight strips. Although DCC, HHM, and LORACCS demonstrate better performance in global color consistency, DCC suffers from severe overall color casts, whereas HHM and LORACCS provide only limited improvement in this regard. Compared with these methods, the proposed method achieves optimal performance in eliminating color casts, enhancing local uniformity, and ensuring mosaicking continuity, realizing the unification of color balance and texture preservation. The superior performance of the proposed method on the DMZ1 dataset can be attributed to the explicit modeling of radiometric distortion caused by illumination geometry. By comparison, purely statistical or data-driven methods lack such physical constraints and therefore tend to either undercorrect or introduce color distortion.

The primary characteristic of the DMZ2 dataset is the significant variation in illumination conditions, with prominent local brightness nonuniformity, particularly making it prone to significant color discrepancies at flight-strip intersections.

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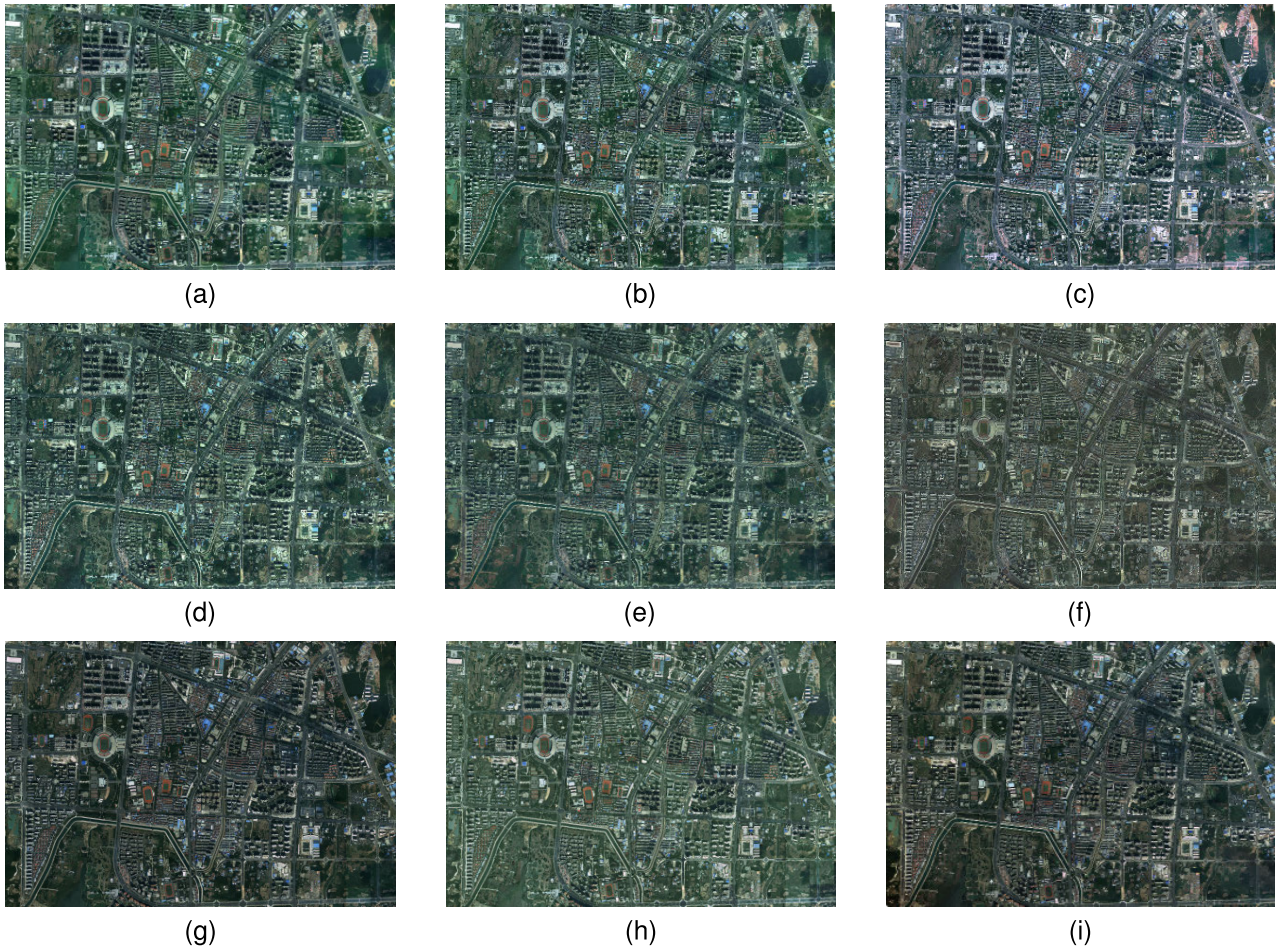


Fig. 7. Mosaicking results of all images for each method on the DMZ1 dataset. (a) Original. (b) Wallis. (c) Retinex. (d) HHM. (e) LORACCS. (f) DCC. (g) C4RSD. (h) RS-NormGAN. (i) Ours.

Thus, this dataset is well suited for assessing the stability of different methods under complex lighting conditions. As shown in Fig. 9, DCC, C4RSD, and the proposed method effectively mitigate local brightness nonuniformity. However, DCC results in weakened details and color distortions, while the Wallis, Retinex, HHM, and LORACCS algorithms fail to effectively eliminate local brightness nonuniformity, and the correction effect of RS-NormGAN remains marginal and unapparent. This intraimage radiometric imbalance is clearly visible from the inconsistent soil brightness on the left and right sides of the orange bounding boxes in the single-image correction results of Fig. 9. From Figs. 10 and 11, it is evident that DCC produces significant detail loss. Regarding mosaicking results, the Wallis, Retinex, C4RSD, and RS-NormGAN algorithms exhibit noticeable color discrepancies in seam regions, failing to achieve smooth transitions between adjacent images. This boundary color discontinuity is particularly prominent across the mosaicking seamline within the orange bounding boxes in Fig. 11, where sharp radiometric steps remain highly visible. Although DCC, HHM, and LORACCS improve global color consistency, slight color discrepancies remain local regions. By contrast, the proposed method achieves the best overall performance in color consistency, brightness uniformity, and detail preservation, demonstrating particular stability in high-contrast regions. The improved performance of the proposed method on the DMZ2 dataset is closely related to the explicit incorporation of BRDF kernels, which account for variations in illumination and viewing geometry. In high-contrast regions, such as flight-strip intersections, statistical or data-driven methods tend to misinterpret geometry-induced brightness variations as radiometric inconsistencies, leading to residual color discrepancies. By contrast, the proposed method distinguishes physical illumination effects from true radiometric discrepancies, resulting in more stable correction in these challenging areas.

Experimental results across the three datasets indicate that traditional empirical methods (Wallis) possess a certain capability in eliminating local brightness nonuniformity but are often accompanied by color shifts. The image enhancement-based Retinex method is superior in improving visual brightness but is prone to overenhancement. Crucially, recent deep learning-based approaches (DCC, C4RSD, and RS-NormGAN) are heavily restricted by their data-driven nature and training dependency; they usually mandate strictly coregistered, high-resolution true reference imagery to establish paired optimization targets. In the absence of such high-fidelity, high-resolution reference maps, or when encountering unmodeled heterogeneous landscapes, these networks are highly susceptible to severe global color distortions, unnatural color casts, or blurring artifacts of structural details.

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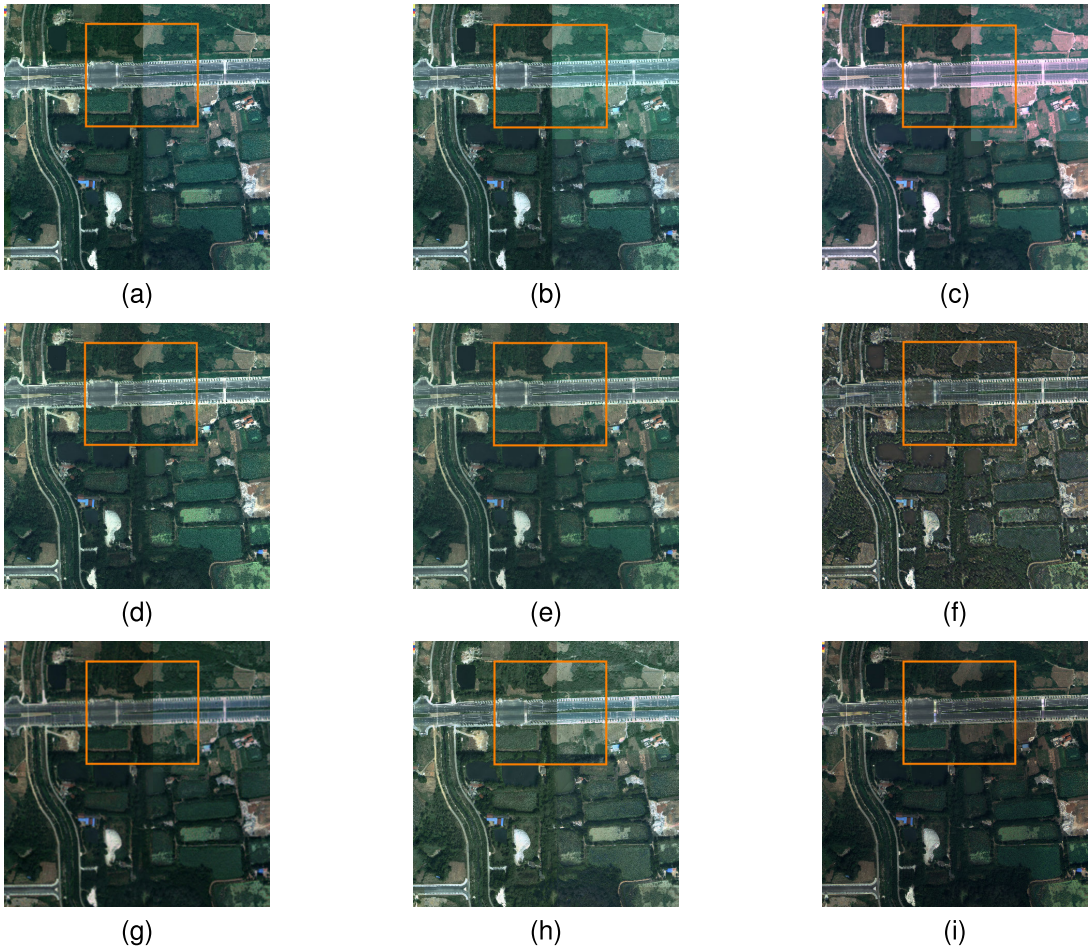


Fig. 8. Local details of mosaicking results for each method on the DMZ1 dataset. (a) Original. (b) Wallis. (c) Retinex. (d) HHM. (e) LORACCS. (f) DCC. (g) C4RSD. (h) RS-NormGAN. (i) Ours.

TABLE II

QUANTITATIVE EVALUATION RESULTS OF DIFFERENT METHODS. BOLD AND UNDERLINED VALUES INDICATE THE BEST AND SECOND-BEST PERFORMANCE, RESPECTIVELY

Method	DMC				DMZ1				DMZ2			
	CD↓	MSE↓	PSNR↑	DeltaE↓	CD↓	MSE↓	PSNR↑	DeltaE↓	CD↓	MSE↓	PSNR↑	DeltaE↓
Original	0.29	297.62	23.60	9.32	1.10	870.44	19.06	7.38	0.69	284.08	23.76	6.81
Wallis [5]	0.98	273.75	23.95	8.84	1.15	808.00	19.37	7.00	1.08	306.81	23.37	6.68
Retinex [9]	0.22	197.45	25.39	10.04	1.03	790.48	19.43	8.01	0.66	262.93	24.14	7.16
HHM [31]	0.50	347.99	23.40	9.41	0.89	673.34	20.24	6.89	0.78	268.60	23.95	6.57
LORACCS [34]	0.45	351.53	23.03	8.21	0.90	511.94	21.40	6.25	0.84	234.22	24.52	6.05
DCC [26]	0.22	**168.79**	**25.89**	7.18	0.69	588.81	20.67	7.13	**0.40**	300.90	23.38	6.99
C4RSD [28]	0.61	288.63	25.30	6.71	0.75	738.12	21.06	6.53	0.61	388.47	24.73	6.87
RS-NormGAN [25]	0.31	345.52	22.94	8.77	0.62	760.62	19.56	7.51	0.60	449.24	23.87	6.61
Ours	**0.14**	315.59	23.58	**4.77**	**0.57**	**502.17**	**21.61**	**4.48**	0.59	**213.69**	**25.10**	**5.50**
Gain vs. Original (%)	*+51.7%*	*-6.0%*	*-0.1%*	*+48.8%*	*+48.2%*	*+42.3%*	*+13.4%*	*+39.3%*	*+14.5%*	*+24.8%*	*+5.6%*	*+19.2%*

Furthermore, radiometric and color balancing methods based on overlapping region constraints (HHM and LORACCS) can improve global consistency to a certain degree but still struggle to simultaneously address intraimage nonuniformity and interimage consistency. In comparison, the proposed method achieves superior performance in both single-image color balancing and multi-image global consistency, while effectively preserving surface texture details. A prominent engineering advantage of the proposed method is its highly flexible data requirement: unlike deep learning models that dictate costly, paired high-resolution reference maps, our framework operates in a completely training-free manner and merely requires easily accessible, low-resolution satellite imagery (e.g., Landsat or Sentinel) as macro global radiometric control anchors. By seamlessly embedding these low-resolution sparse RCPs into the global joint block-adjustment network, the proposed pipeline delivers physics-consistent radiometric harmony across multistrip image blocks, drastically minimizing data-acquisition overheads for large-scale photogrammetric production.

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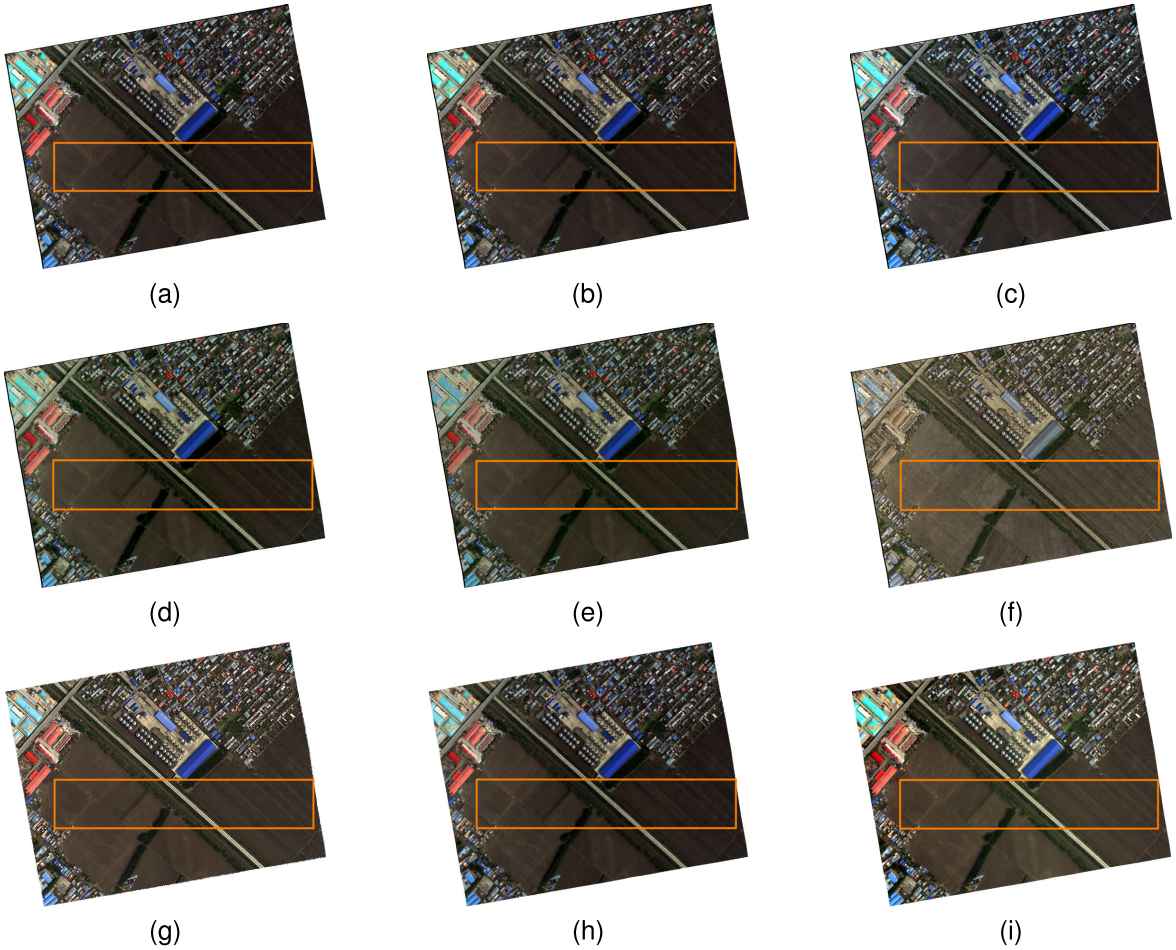


Fig. 9. Single-image color correction results for each method on the DMZ2 dataset. (a) Original. (b) Wallis. (c) Retinex. (d) HHM. (e) LORACCS. (f) DCC. (g) C4RSD. (h) RS-NormGAN. (i) Ours.

C. Quantitative Evaluation

The quantitative performance of various methods across the three datasets was evaluated using CD, mse, PSNR, and DeltaE. The experimental results are summarized in Table II, where each value represents the average over the corresponding dataset. As evidenced by the results, the proposed method achieves the best performance on most evaluation metrics across all three datasets. The advantages are particularly pronounced on the DMZ1 and DMZ2 datasets, which are characterized by uneven illumination and severe color casts. For the DMC dataset, due to the high image quality and minimal interimage discrepancies, all methods maintain stable performance with PSNR exceeding 22 dB. Nevertheless, the proposed method still achieves the best performance in terms of DeltaE, which reflects intraimage color balance, and CD, which measures interimage color consistency. This reveals its noticeable advantage in preserving true colors and avoiding overcorrection. For the DMZ1 dataset, the proposed algorithm achieves optimal performance on all four evaluation metrics. This result highlights the synergistic advantages of the physical radiometric model and the radiometric aerotriangulation consistency correction model in handling scenarios with significant illumination variations and severe color casts. In particular, the substantial reduction in DeltaE stems from the capability of the physical radiometric model to compensate for local radiometric nonuniformity at the imaging mechanism level, resulting in more natural and consistent colors within individual images. Regarding the DMZ2 dataset, the proposed method maintains its leading performance in radiometric realism, capturing the lowest mse (213.69) and the highest PSNR (25.10). Although DCC shows a slight advantage in the CD metric, this localized gain is achieved at the catastrophic expense of massive absolute pixel distortions (mse elevates to 300.90) and severe detail loss. This numeric decoupling validates that purely data-driven deep transformations lack rigid physical boundaries, regularly generating uninterpretable color bleeding. In contrast, the proposed method demonstrates superior consistency across mse, PSNR, and DeltaE metrics, reflecting a more balanced improvement in global consistency, radiometric realism, and detail preservation. As illustrated in Table II, the proposed algorithm exhibits the best comprehensive performance in radiometric and color balancing for aerial imagery, offering distinct improvements particularly when dealing with lower quality imagery.

It is worth noting that improvements in different metrics reflect different aspects of color consistency. While some comparative methods achieve relatively low CD values, this improvement is often accompanied by increased DeltaE or visual color distortion, indicating a tradeoff between

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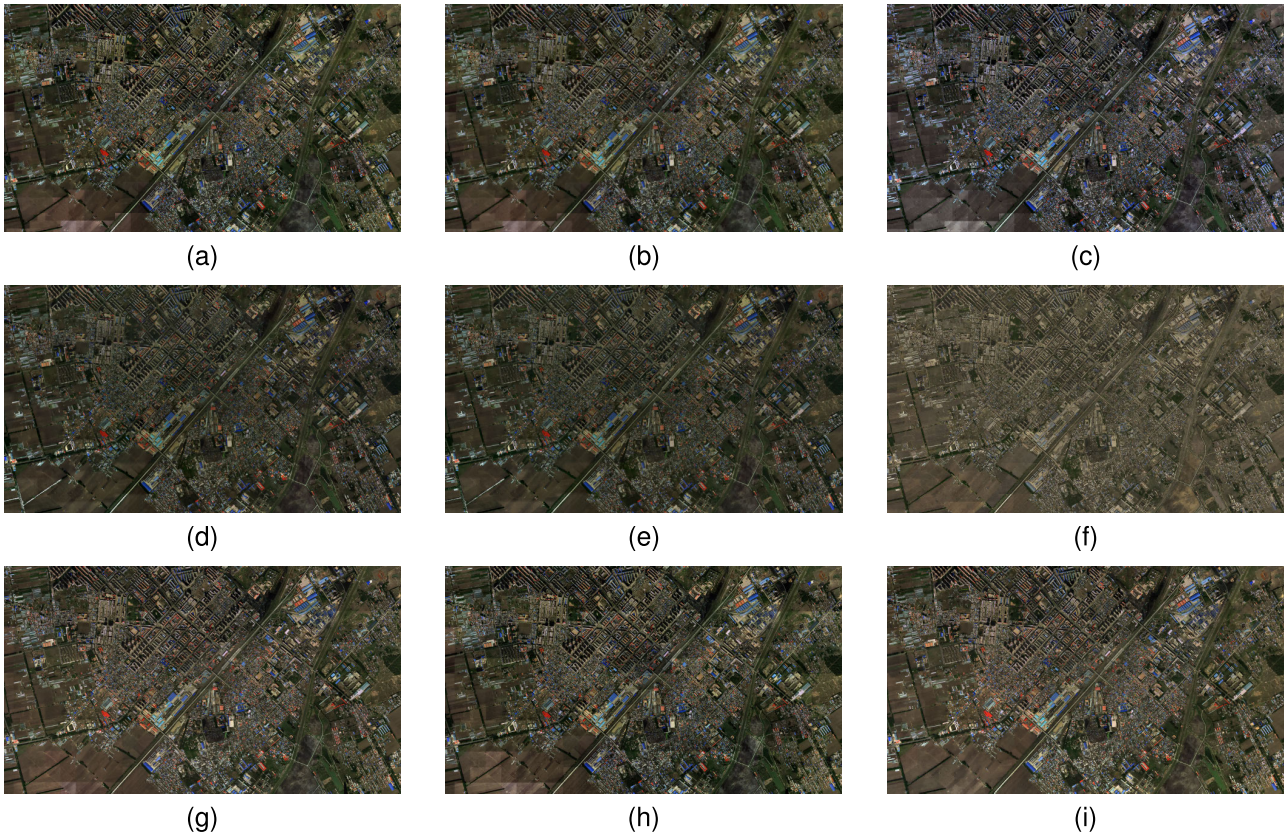


Fig. 10. Mosaicking results of all images for each method on the DMZ2 dataset. (a) Original. (b) Wallis. (c) Retinex. (d) HHM. (e) LORACCS. (f) DCC. (g) C4RSD. (h) RS-NormGAN. (i) Ours.

TABLE III

COMPUTATIONAL EFFICIENCY AND HARDWARE OPERATIONAL COST COMPARISON OF DIFFERENT METHODS. BOLD AND UNDERLINED VALUES INDICATE THE BEST AND SECOND-BEST PERFORMANCE, RESPECTIVELY

Method	DMC			DMZ1			DMZ2		
	Total Time (h)	Single Image (min)	FLOPs (G)	Total Time (h)	Single Image (min)	FLOPs (G)	Total Time (h)	Single Image (min)	FLOPs (G)
Wallis [5]	<u>6.18</u>	<u>0.34</u>	**0.64**	**3.27**	**0.35**	**1.22**	**1.69**	**0.36**	**1.14**
Retinex [9]	151.47	8.29	91.34	79.38	8.47	175.31	41.75	8.76	164.05
HHM [31]	29.03	1.59	1.59	15.24	1.63	3.05	7.31	1.53	2.85
LORACCS [34]	38.99	2.13	1.67	30.87	3.30	3.20	13.12	2.75	3.00
DCC [26]	18.04	0.99	1.04×10^5	25.38	2.71	1.92×10^5	13.90	2.92	1.82×10^5
C4RSD [28]	16.74	0.92	6.13×10^4	13.72	1.47	1.13×10^5	5.08	1.07	1.07×10^5
RS-NormGAN [25]	6.28	<u>0.34</u>	9.26×10^4	5.21	0.56	1.71×10^5	2.68	0.56	1.62×10^5
Ours	**5.32**	**0.29**	<u>1.30</u>	<u>4.47</u>	<u>0.48</u>	<u>2.49</u>	<u>2.56</u>	<u>0.54</u>	<u>2.33</u>

interimage consistency and intraimage color fidelity. By contrast, the proposed method achieves balanced improvements across CD, mse, PSNR, and DeltaE, demonstrating its ability to jointly optimize global consistency and local color uniformity under physical constraints.

The experimental results in Table II indicate that the proposed method outperforms the comparative methods in terms of physical consistency, visual naturalness, and interimage color balance. The proposed method achieves the lowest DeltaE on all three datasets, signifying a substantial enhancement in intraimage color balance. Moreover, it exhibits the most stable and balanced comprehensive performance across the global consistency metrics (CD, mse, and PSNR). While purely data-driven models like DCC occasionally present localized numerical advantages in isolated metrics due to aggressive local pixel mappings or generalized tone smoothing, they frequently introduce severe coupled spectral distortions elsewhere. In contrast, by decoupling physical illumination compensation from block-adjustment structures, the proposed framework simultaneously ensures local and global color consistency under complex illumination conditions.

D. Computational Efficiency and Cost Analysis

Table III reports the quantitative evaluation of computational efficiency across the three distinct datasets. To ensure a rigorous and fair comparison, all computational tests were executed on a standardized workstation equipped with an Intel Core CPU, 64 GB of RAM, an NVIDIA GeForce RTX 3060 (12-GB VRAM) GPU, and a 7-TB solid-state drive (SSD). Crucially, all CPU-based algorithms were limited to a strict single-thread configuration to eliminate hardware parallelism bias.

As evidenced by the results, the proposed framework achieves an outstanding balance between theoretical computational frugality and actual processing speed. Purely empirical

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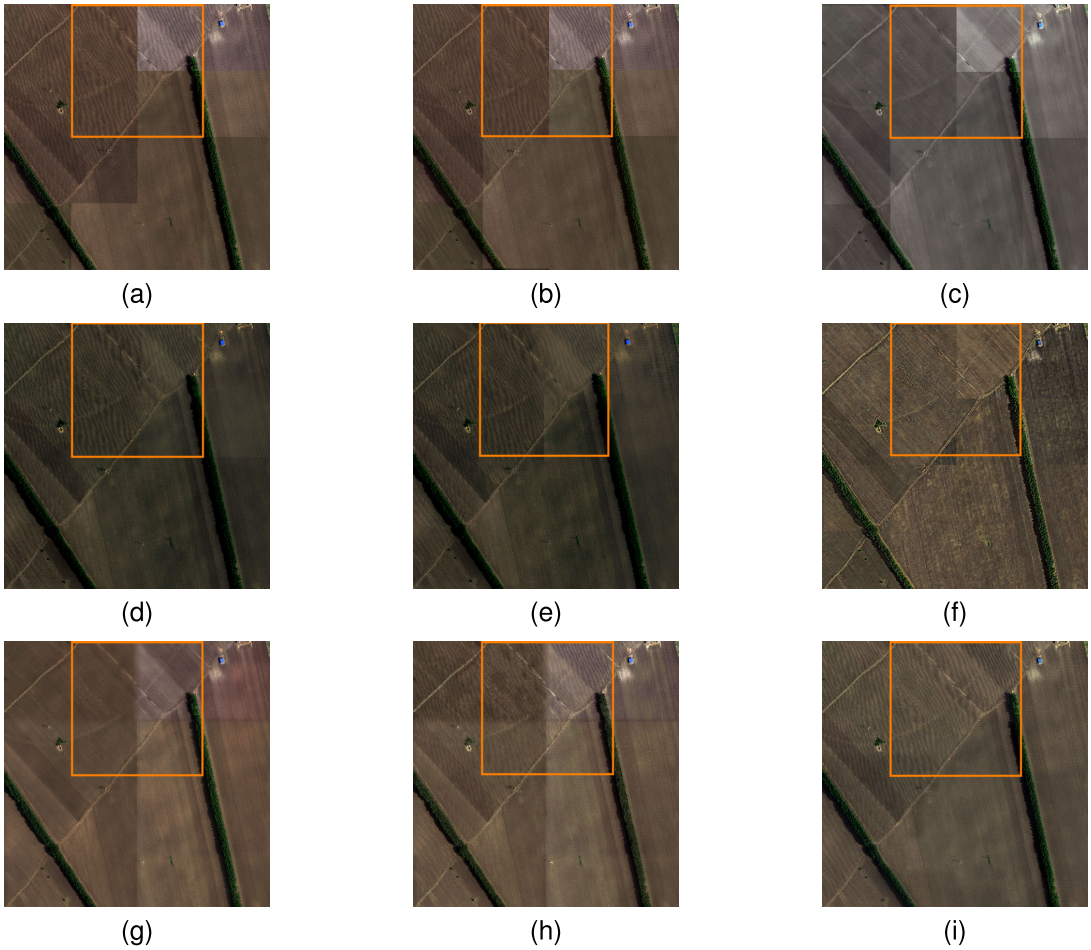


Fig. 11. Local details of mosaicking results for each method on the DMZ2 dataset. (a) Original. (b) Wallis. (c) Retinex. (d) HHM. (e) LORACCS. (f) DCC. (g) C4RSD. (h) RS-NormGAN. (i) Ours.

local filters, such as the Wallis algorithm, maintain the lowest FLOPs constraints (approximately 0.64–1.22 G). However, as previously demonstrated in the qualitative analysis, this statistical clipping fails to structurally eliminate severe radiometric nonuniformities. In stark contrast, recent deep learning approaches (DCC, C4RSD, and RS-NormGAN) suffer from an immense computational penalty, where their layered tensor convolutions and dense cross-attention matrices elevate the operational workloads to colossal levels exceeding 1.0×10^5 GFLOPs. Although these data-driven networks leverage high-end GPU acceleration architectures to shorten their single-image inference latency (e.g., RS-NormGAN achieving 0.34–0.56 min per image), their total block processing time remains uncompetitive when subjected to heavy tile-partitioning and high-resolution data IO configurations. Operating entirely on a standard CPU pipeline, the proposed method strategically circumvents these hardware-intensive limits. By resolving joint consistency equations using sparse matrix Cholesky factorization, our framework limits the pixel-scale computational cost to a mere 1.30–2.49 GFLOPs. Consequently, it achieves the shortest total processing time on the large-scale DMC dataset (5.32 h) and consistently maintains highly competitive runtime performance across the DMZ1 and DMZ2 datasets, securing prominent operational feasibility for large-scale production.

In summary, the comparative analysis of computational cost self-evidently establishes the engineering superiority and operational feasibility of the proposed physically grounded radiometric aerotriangulation. While image enhancement-based alternatives like Retinex introduce severe computational bottlenecks (requiring up to 151.47 h on DMC) and deep learning pipelines demand expensive hardware clusters along with rigorous data-pretraining latencies, the proposed complete framework delivers a lightweight, zero-shot, and hardware-agnostic solution. By coupling highly compressed physical modeling with robust sparse global optimization, our system guarantees deterministic, high-fidelity color consistency with minimal computational overhead, proving highly scalable and immediately ready for commercial industrial deployments.

E. Ablation Study

To validate the individual contributions and synergistic effects of the distinct modules in the proposed radiometric and color balancing method for aerial imagery based on remote sensing radiometric references, a systematic ablation study was conducted. By comparing correction results across different model combinations, the respective roles of these modules in achieving image color consistency were analyzed.

1) Experimental Setup: To comprehensively evaluate the robustness and effectiveness of the proposed method under

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Fig. 12. Comparison of radiometric and color balancing results obtained under different configurations for the DMC dataset. (a) Configuration (1). (b) Configuration (2). (c) Configuration (3). (d) Configuration (4). (e) Configuration (5). In each column, the images from top to bottom represent single-image correction result, global mosaicking result, and local mosaicking details.

varying initial conditions, datasets DMC, DMZ1, and DMZ2 were selected for the ablation and comparative experiments. The overall framework of the proposed method comprises two core modules: the physical radiometric model for nonuniformity color compensation and the color consistency correction model based on radiometric aerotriangulation. To evaluate the independent and joint effects of these modules, five experimental configurations were designed: 1) direct mosaicking using original images; 2) radiometric and color balancing using the HHM algorithm [31]; 3) color correction using only the physical radiometric model for nonuniformity color compensation; 4) color correction using only the consistency correction model based on radiometric aerotriangulation; and 5) the proposed complete method. Specifically, HHM in configuration 20 represents a classic empirical color correction approach widely used in aerial photogrammetry production. It establishes hybrid statistical transfer functions by balancing global color distribution characteristics with localized constraints derived from overlapping image regions. Since such statistical approaches do not rely on physical imaging geometry or structural aerotriangulation networks, they serve as a standard baseline to evaluate the performance margins between purely data-driven statistical matching and the proposed physically constrained global optimization pipeline.

2) Evaluation Metrics: DeltaE is employed to assess the degree of intraimage color balance, while CD, mse, and PSNR are used to measure interimage color consistency.

Fig. 12 presents a visual comparison of radiometric and color balancing results obtained under different configurations for the DMC dataset, including single-image correction results, global mosaicking results, and local mosaicking details. The evaluation on the DMC dataset provides a critical perspective on the system's stability. Unlike DMZ1 and DMZ2, the original DMC imagery possesses superior initial radiometric quality with minimal severe color casts. A common limitation of aggressive color compensation algorithms is the tendency to overcorrect high-quality images, often introducing unnatural brightness or washing out fine textures. However, as shown in the local mosaicking details of Fig. 12(a) and (d), the proposed algorithm effectively fine-tunes the subtle interimage discrepancies while strictly preserving the intrinsic high-fidelity textures and dynamic range of the original DMC data. This completely avoids the overcorrection phenomenon, proving that the proposed physically based radiometric aerotriangulation is highly stable and acts as a safe, unified pipeline for datasets of varying qualities.

Furthermore, Fig. 13 presents the experimental results for the DMZ1 dataset, which is characterized by significant illumination nonuniformity and severe color discrepancies. As shown in the figure, the original images (1) exhibit uneven internal brightness distribution, with color discrepancies between central and edge regions, leading to color discontinuities near mosaicking boundaries and poor overall visual quality. The traditional method (2) utilizes information from overlapping areas between images to achieve color transitions through local optimization, leading to a slight improvement in global color consistency. Nevertheless, intraimage brightness nonuniformity persists, resulting in local color casts in the mosaicking results. When only the physical radiometric model is applied (3), intraimage color balance improves significantly, and edge details are recovered. However, due to the lack of cross-image constraints, color discrepancies remain between different flight strips, resulting in visible color discrepancies across seamlines. When using only the consistency correction model based on radiometric aerotriangulation (4), global brightness becomes more uniform and mosaicking continuity is substantially improved. Nevertheless, the issue of intraimage brightness nonuniformity remains unaddressed, with local regions exhibiting brightness inconsistencies. The complete method (5), which integrates local physical compensation with global radiometric constraints, achieves natural overall color, smooth transitions at

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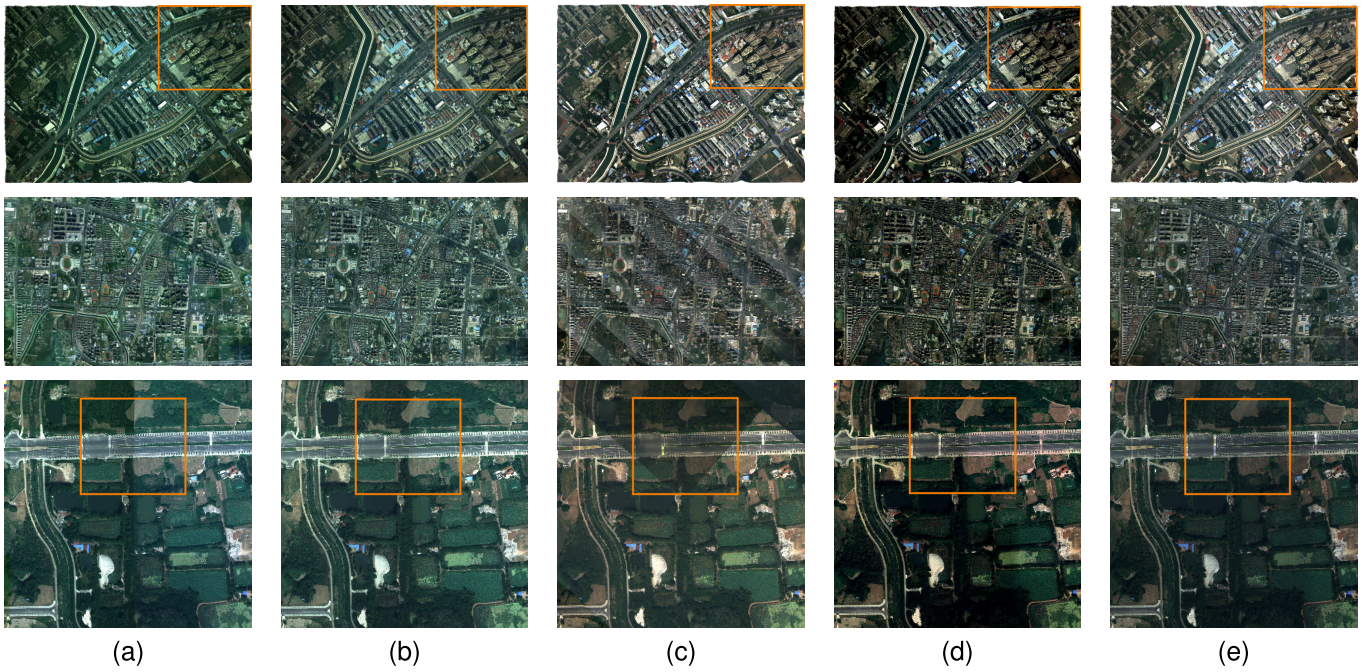


Fig. 13. Comparison of radiometric and color balancing results obtained under different configurations for the DMZ1 dataset. (a) Configuration (1). (b) Configuration (2). (c) Configuration (3). (d) Configuration (4). (e) Configuration (5). In each column, the images from top to bottom represent single-image correction result, global mosaicking result, and local mosaicking details.

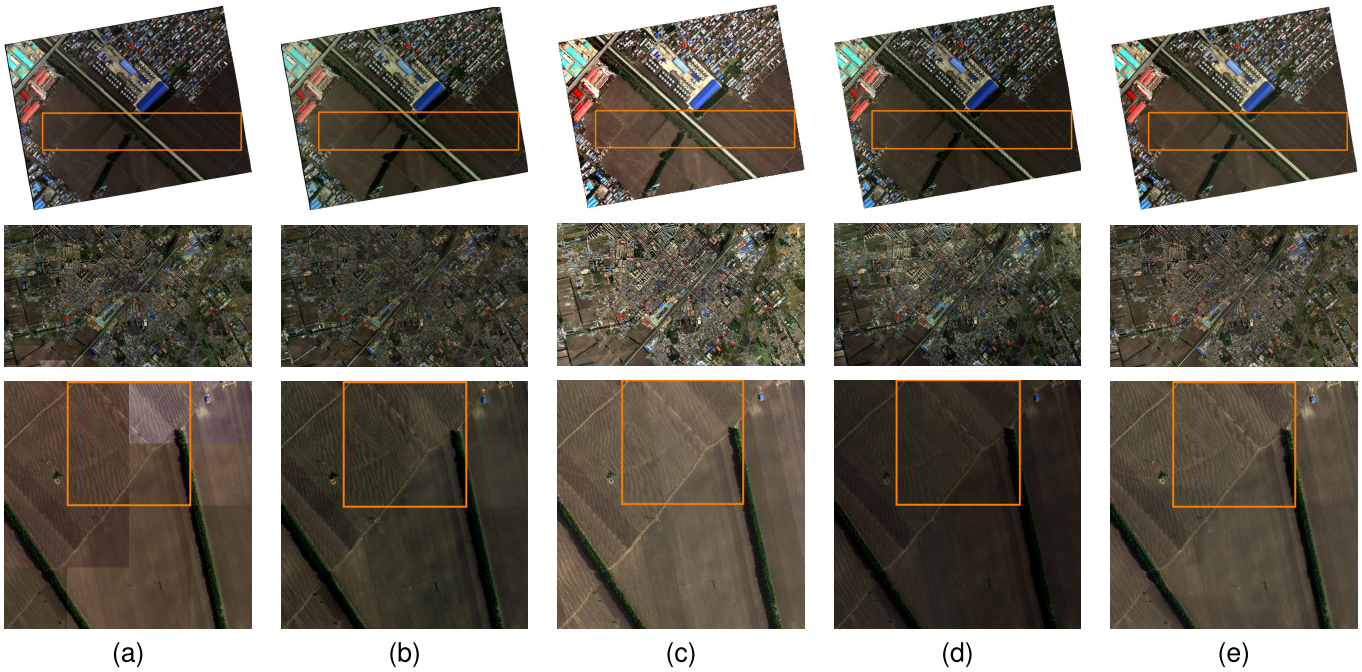


Fig. 14. Comparison of radiometric and color balancing results obtained under different configurations for the DMZ2 dataset. (a) Configuration (1). (b) Configuration (2). (c) Configuration (3). (d) Configuration (4). (e) Configuration (5). In each column, the images from top to bottom represent: single-image correction result, global mosaicking result, and local mosaicking details, respectively.

seam regions, and excellent preservation of texture details. It delivers the best performance in both visual and physical consistency. The experimental results indicate that the proposed method effectively balances local detail fidelity and global color consistency, achieving high-quality radiometric and color balancing for aerial imagery. Local magnification results further indicate that the complete method maintains distinct detail hierarchies and color consistency even in high-contrast areas such as roads and shadow boundaries, reflecting the physical consistency and radiometric correction stability of the model.

Finally, Fig. 14 displays the experimental results for the DMZ2 dataset, demonstrating the algorithm's robust generalization across different topographical features and varying illumination conditions. The DMZ2 dataset is primarily characterized by significant fluctuations in lighting and prominent local brightness nonuniformity, which frequently lead to noticeable color discrepancies, particularly at flight-strip

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TABLE IV

QUANTITATIVE EVALUATION RESULTS OF DIFFERENT CONFIGURATIONS IN THE ABLATION STUDY. BOLD AND UNDERLINED VALUES INDICATE THE BEST AND SECOND-BEST PERFORMANCE, RESPECTIVELY

Configuration	DMC				DMZ1				DMZ2			
	CD↓	MSE↓	PSNR↑	DeltaE↓	CD↓	MSE↓	PSNR↑	DeltaE↓	CD↓	MSE↓	PSNR↑	DeltaE↓
(1)	<u>0.29</u>	**297.62**	**23.60**	9.32	1.10	870.44	19.06	7.38	0.69	284.08	23.76	6.81
(2)	0.50	347.99	23.40	9.41	0.89	<u>673.34</u>	<u>20.24</u>	6.89	0.78	268.60	23.95	6.57
(3)	0.97	621.32	20.42	<u>5.48</u>	0.76	887.49	19.05	<u>5.30</u>	**0.45**	398.87	22.36	<u>6.46</u>
(4)	0.42	595.56	20.62	6.52	<u>0.61</u>	685.91	20.16	6.31	<u>0.54</u>	<u>241.74</u>	<u>24.76</u>	7.44
(5)	**0.14**	<u>315.59</u>	<u>23.58</u>	**4.77**	**0.57**	**502.17**	**21.61**	**4.48**	0.59	**213.69**	**25.10**	**5.50**
Gain vs. Original (%)	*+51.7%*	*-6.0%*	*-0.1%*	*+48.8%*	*+48.2%*	*+42.3%*	*+13.4%*	*+39.3%*	*+14.5%*	*+24.8%*	*+5.6%*	*+19.2%*

intersections. As observed in the original imagery [Fig. 14(a)], the intraimage brightness is highly uneven, clearly evident from the inconsistent soil brightness on the left and right sides of the orange bounding box in the single-image correction result. Furthermore, color discontinuities are prominent near the mosaicking boundaries between flight strips, as highlighted in the local mosaicking details. The traditional statistical method [Fig. 14(b)] smooths transitions but leaves intrinsic intraimage nonuniformity unaddressed. Similarly, applying only the physical radiometric model [Fig. 14(c)] improves intraimage balance but leaves slight discrepancies between flight strips, whereas relying solely on radiometric aerotriangulation [Fig. 14(d)] enhances boundary continuity but fails to eliminate the visible striping effects in the global mosaic caused by vignetting. In contrast, the proposed complete method [Fig. 14(e)] seamlessly integrates local physical compensation with global radiometric constraints. It achieves a natural overall color tone and smooth boundary transitions while strictly preserving high-frequency texture details, successfully handling the severe cross-strip inconsistencies. These results systematically confirm that the proposed framework optimally balances local detail fidelity with global color consistency, delivering high-quality radiometric balancing even under challenging and variable illumination conditions.

Table IV presents the quantitative evaluation results of different configurations across the three datasets (DMC, DMZ1, and DMZ2). As shown in the table, the quantitative trends consistently validate the synergistic effectiveness of the proposed modules under varying initial conditions. For the highly degraded DMZ1 dataset, configuration (1) exhibits poor performance characterized by substantial brightness fluctuations and severe color imbalance. Although the traditional method (2) improves mosaicking continuity and global color consistency to a certain extent, intraimage brightness nonuniformity persists. Compared with the traditional method (2), configuration (3) achieves reductions of approximately 15% and 23% in CD and DeltaE, respectively. This demonstrates that physical radiometric modeling can effectively compensate for local illumination variations, thereby mitigating local brightness nonuniformity and color casts. Nevertheless, due to the absence of global constraints across multiple images, brightness discrepancies persist in mosaicking regions. When only the consistency correction model based on radiometric aerotriangulation (4) is applied, CD reduces to 0.61. This indicates that introducing radiometric constraints within the aerotriangulation framework can effectively suppress residual interimage color discrepancies and enhance global consistency. However, since this approach employs a unified linear model for each image, its ability to recover intraimage details is limited, leaving the issue of intraimage brightness nonuniformity unresolved. The complete configuration (5) achieves the optimal performance balance. Taking the DMZ1 dataset as an example, compared with the original images, CD, mse, and DeltaE decreased by 48%, 42%, and 39%, respectively, while PSNR improved by 13%. These synergistic effects are further corroborated by the DMZ2 dataset, where the complete method (5) consistently secures the lowest mse and highest PSNR, demonstrating robust generalization under different topographical and illumination conditions. Most importantly, the quantitative metrics for the high-quality DMC dataset strongly support the system's stability. Unlike aggressive empirical or local smoothing approaches that frequently overcorrect and distort high-quality baseline data, the proposed complete method (5) safely refines the imagery. It drastically reduces DeltaE (from 9.32 to 4.77) and CD (from 0.29 to 0.14) while maintaining a high and stable PSNR (23.58). This quantitatively proves that the proposed framework successfully avoids the overcorrection phenomenon, strictly preserving the original high-fidelity structural data while achieving seamless global color consistency.

The ablation study results indicate that the physical radiometric model (3) successfully improves intraimage nonuniformity, leading to more balanced brightness and color distributions, but brightness discrepancies between flight strips remain. The consistency correction model based on radiometric aerotriangulation (4) establishes radiometric constraints across multiple images, effectively reducing color discrepancies between adjacent images and flight strips, yet the issue of intraimage brightness nonuniformity persists. The complete method (5) integrates the advantages of both modules and achieves the best performance in terms of both local uniformity and global consistency. Specifically, a strong synergistic effect exists between these two components. On the one hand, the physical model effectively eliminates intraimage vignetting and BRDF effects, thereby providing unbiased and highly consistent tie-point observations for the subsequent global adjustment. Without this, the raw tie points would carry position-dependent illumination errors, severely degrading the adjustment accuracy. On the other hand, the radiometric aerotriangulation provides the optimal global scaling and baseline

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Fig. 15. Typical failure cases observed in (a) DMC, (b) DMZ1, and (c) DMZ2 datasets. The orange bounding boxes highlight localized radiometric discrepancies over water bodies near the mosaicking seamlines, primarily caused by specular reflection and sun glint variations.

parameters (s_i and a_{0i}) to seamlessly anchor these locally flattened images to the absolute Sentinel-2 reference. These findings demonstrate that neither local compensation nor global optimization alone is sufficient to address the prominent issues of intraimage illumination nonuniformity and interimage color discrepancies in aerial imagery. Their synergistic integration provides an effective and comprehensive solution to these challenges.

F. Discussion

Experimental results indicate that the proposed radiometric and color balancing method outperforms comparative methods across three representative aerial datasets. Overall, the proposed method exhibits advantages in two primary aspects.

1) The physical radiometric model effectively suppresses intraimage nonuniformity. In single-image correction, traditional methods predominantly rely on local filtering or histogram adjustments, which are essentially statistical equalization processes and therefore prone to oversmoothing or color shifts. By comparison, the proposed physical radiometric model simulates radiative transfer and sensor response during the imaging process, providing a clear physical basis for color correction. This enables the elimination of brightness nonuniformity while preserving texture and edge details.
2) The color consistency correction model based on radiometric aerotriangulation enhances global consistency across multiple images. Traditional color balancing methods based on overlapping regions rely solely on local matching and lack global constraints, which often leads to accumulated color discrepancies between flight strips. By integrating radiometric parameters into the aerotriangulation framework, the proposed model achieves a unified optimization of radiometric consistency and geometric positioning. This integrated framework successfully eliminates interstrip discrepancies and color differences in overlapping areas, resulting in smooth and natural mosaicking boundaries.

Despite the favorable performance demonstrated across multiple datasets, certain limitations remain and warrant further investigation. First, the proposed physical radiometric model primarily compensates for color discrepancies caused by uneven illumination and variations in imaging viewing angles. It cannot fully recover missing information resulting from complex interactions between scene geometry and illumination, such as cast shadows, self-shadows, and highly reflective targets.

To further elucidate the boundaries of our framework, specific failure cases from the DMC, DMZ1, and DMZ2 datasets are detailed in Fig. 15. As highlighted by the orange bounding boxes across all three datasets, distinct color discrepancies persist at the seamlines over localized water bodies. This phenomenon stems from the intrinsic limitation of the physically based empirical model when dealing with specular surfaces. The implemented RTLSR kernel combination is theoretically grounded on volumetric and geometric-optical scattering, which ideally fits diffuse terrestrial surfaces such as vegetation and soil. However, water bodies are characterized by dominant specular reflection and severe sun glint. Consequently, even a minor variation in the solar-target-sensor geometry between adjacent flight strips can lead to catastrophic brightness divergence over water—where one sensor captures intense directional sunglint while the neighboring one records the dark, absorptive intrinsic water color. Furthermore, the dynamic nature of water surfaces (e.g., wind-induced wave facets) disrupts the temporal and spatial stationarity assumed by the radiometric aerotriangulation, causing localized failures in global consistency optimization.

Second, the color consistency correction model based on radiometric aerotriangulation assumes that images are comparable—that is, acquired using similar sensors under proximate temporal and atmospheric conditions. For multisource, multitemporal, or multiseason datasets, the foundational assumptions for unified optimization may be compromised due to significant discrepancies in radiometric references and inconsistent illumination conditions. Consequently, the adaptability of the proposed method to multisource heterogeneous data requires further verification. Furthermore, the method relies on the reliability of radiometric references. If the calibration quality of the reference image is insufficient, or if the distribution of control areas is uneven or occluded, the stability of radiometric constraints and the reliability of the final results may be adversely affected.

Overall, while the proposed method demonstrates robust performance under typical aerial imaging conditions, further research is required to incorporate adaptive specular reflection models and water masking mechanisms to extend its applicability to more complex and diverse operational scenarios.

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V. Conclusion

To address the issues of intraimage illumination nonuniformity and significant interimage color discrepancies in aerial imagery caused by factors such as variations in illumination conditions, terrain undulation, and nonlinear sensor responses, this article proposes a radiometric and color balancing method based on remote sensing radiometric references. This method innovatively integrates a physical radiometric model with the aerotriangulation framework, providing a unified solution for both intraimage nonuniformity compensation and multiimage global consistency correction. Experimental evaluations were conducted on three representative aerial datasets. Comparative analyses were performed against several established techniques, encompassing the Wallis algorithm, Retinex algorithm, HHM, LORACCS algorithm, and deep learning-based approaches (DCC, C4RSD, and RS-NormGAN). The empirical results indicate that the proposed method outperforms comparative methods across the majority of evaluation metrics, exhibiting superior performance particularly in complex scenarios characterized by uneven illumination and significant color casts. Specifically, relative to the top-performing benchmark, the proposed method achieved reductions in CD and DeltaE of 36.36% and 28.91%, respectively, on the DMC dataset. For the DMZ1 dataset, which exhibits severe color casts, CD, mse, and DeltaE decreased by 8.06%, 1.91%, and 28.32%, respectively, while PSNR increased by 0.21 dB. In the DMZ2 dataset, characterized by large illumination differences, mse and DeltaE were reduced by 8.77% and 9.09%, respectively, with a PSNR improvement of 0.37 dB. Notably, the method exhibits superior performance in terms of enhancing color consistency, preserving radiometric fidelity, and retaining texture details. Consequently, it provides an effective and robust solution for elevating the quality of remote sensing products.

Future work will focus on the following three aspects: first, further improving the computational efficiency of the proposed sparse block-adjustment optimization framework for massive, national-scale aerial datasets; second, incorporating semantic segmentation or land-cover classification priors into the pipeline to drive surface-specific, sophisticated BRDF kernels, thereby eliminating localized color biases over nonvegetated or specular surfaces (such as water bodies and artificial manmade structures); and third, expanding the potential application of the method to multisource and multitemporal remote sensing imagery, thereby providing more reliable data support for high-precision geospatial analysis and change detection.

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Yuting Qin received the B.E. degree in remote sensing science and technology from Wuhan University, Wuhan, China, in 2022, where he is currently pursuing the Ph.D. degree.

His research interests include image registration, 3-D reconstruction, and sensing photogrammetry.



Zhiling Yu received the B.E. degree from the School of Remote Sensing and Information Engineering, Wuhan University, Wuhan, China, in June 2025. He is currently pursuing the M.E. degree with Wuhan University.

His research interests include photogrammetry and remote sensing image processing.



Yansong Duan was born in 1975. He received the M.S. and Ph.D. degrees from Wuhan University, Wuhan, China, in 2009 and 2016, respectively.

He is currently an Associate Professor with the School of Remote Sensing and Information Engineering, Wuhan University. His research interests include photogrammetry, image processing and matching, 3-D city reconstruction, computer vision, and high-performance computing.

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